

Data Analysis of the Top 250 Portuguese Companies Dataset

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Abstract. This report presents the complete analysis of the dataset of the Top 250 Portuguese companies by Turnover in 2022. We start with an exploratory univariate and bivariate analysis of the dataset, which provided a general understanding of the variables, the data for each variable, and the potential relations between the variables. Then we conducted a multivariate analysis, firstly using Principal Component Analysis and then Cluster Analysis. We identified, among the 250 largest companies, that there is a group of about 5% that are much larger than the rest. We also found that the companies are geographically concentrated (about 70% are located in Lisbon or Porto), and that the type of economic activity has a statistically significant influence in the financial and economic structure of the companies. In the Principal Component Analysis, we found that EBITDA and financial structure variables are the main attributes through which we can compare and distinguish the companies, followed by number of employees and associated costs, and by sales volume and associated costs. In the cluster analysis, we found that the 250 largest Portuguese companies are composed of a majority of relatively healthy companies, and two smaller groups of economically and financially poor companies, and of exceptionally good companies. Around 80% of the companies are financially and economically healthy.

Keywords: Data Analysis, Exploratory Data Analysis, Principal Component Analysis, Cluster Analysis, Portuguese Companies

1 Introduction

In this report, we will be performing the complete analysis of a dataset composed of financial and economic data of the Top 250 Portuguese companies by Turnover (Total Sales of Goods and Services) in 2022. The data was obtained on October the 15th, 2023 from the SABI Informa database. This is a database that presents economical and financial data about 2.900.000 Spanish companies and 900.000 Portuguese companies and is available at the Library of Faculdade de Economia do Porto.

In the univariate analysis, our aim is to explore the dataset and understand the variables and the quality of the data for each of them. For this purpose, we will be analysing the type of variables, the presence of missing values, and the location, dispersion and shape of the variables, both graphically and through a range of statistical indicators.

As for the bivariate analysis, we aim to find potential relations between the variables. It will be interesting to find whether there is a relationship between the different financial and economical attributes of a company, or between those attributes and the type of economic activity, and whether there is a general pattern or tendency in the economic structure of the Top 250 Portuguese companies. For this purpose, we will analyse graphs and conduct statistical tests.

In the multivariate analysis, we will analyse all variables simultaneously in order to detect interesting patterns, identify the main dimensions and attributes and create

meaningful representations of the largest 250 Portuguese companies. For this purpose, we will implement Principal Component Analysis and Cluster Analysis.

1.1 Data

This dataset is composed of 250 rows, which corresponds to the top 250 companies according to Turnover (represented by the variable “Volume_de_Negocios”), and 20 columns, where the first one, “Nome”, can be understood as the Entity ID, and the remaining 19 are attribute variables. In the following table we present the variables and their respective type:

Type of Variable	Name of the Variable
Entity ID	Nome
Categorical Nominal	Distrito
	CAE_agr
Numerical Continuous	Imobilizado
	Activo_Circulante
	Total_activo
	Capital_Proprio
	Passivo_MLP
	Fundo_de_manejo
	Volume_de_Negocios
	CMVMC
	Custos_com_Pessoal
	Amortizacoes
	EBITDA
	Resultados_correntes
	Result_Liq
	Stock_Turnover_per
	PMR_dias
	PMP_dias
Numerical Discrete	Numero_de_empregados

Variable Transformations

In order to improve the readability and facilitate the analysis of the variables, we took the following steps:

- We transformed the variable “Localidade” into “Distrito”, that is, the City/Town of each company was replaced by the District, thus reducing the number of categories, and facilitating the interpretation of this variable.
- We transformed “CAE” into “CAE_agr” by reducing each code number into its first two digits, which corresponds to the main aggregated classes of economic activity. The analysis of the fully disaggregated CAE number would not be interesting as there would be too many different categories.
- For the creation of the histograms, boxplots and tables, we transformed all numerical variables except “Numero_de_empregados”, “PMR”, “PMP” and “Stock_turnover_per” from euros to thousands of euros, by dividing the columns by 1000. The purpose was to improve the readability of the data.
- We replaced the missing values in “Stock_turnover_per”, that were represented by the string “n.s.”, into blanks, in order to conduct the analysis of this numerical variable.

New Variables

Moreover, further ahead, in the Multivariate Analysis of the dataset, additional variables will be considered, namely 10 new variables (numerical continuous variables) as ratios of the original variables:

- $\text{Racio_FM (Working Fund Ratio)} = \frac{\text{Capital_Proprio} + \text{Passivo_MLP}}{\text{Imobilizado}}$
- $\text{Liquidez (Liquidity ratio)} = \frac{\text{Activo_Circulante}}{\text{Total_activo} - \text{Capital_Proprio} - \text{Passivo_MLP}}$
- $\text{Auto_Financ (Financial Autonomy)} = \frac{\text{Capital_Proprio}}{\text{Total_activo}}$
- $\text{Racio_Endiv (debt to total assets ratio)} = \frac{\text{Total_activo} - \text{Capital_Proprio}}{\text{Total_activo}}$
- $\text{Solvabilidade (Solvency Ratio)} = \frac{\text{Capital_Proprio}}{\text{Total_activo} - \text{Capital_Proprio}}$
- $\text{Debt_EBITDA (Debt to EBITDA Ratio)} = \frac{\text{Total_activo} - \text{Capital_Proprio}}{\text{EBITDA}}$
- $\text{M_EBITDA (EBITDA Margin)} = \frac{\text{EBITDA}}{\text{Volume_de_Negocios}}$
- $\text{Rent_Vnd (Profit Margin)} = \frac{\text{EBITDA} - \text{Amortizacoes}}{\text{Volume_de_Negocios}}$
- $\text{ROI (Return on Investment)} = \frac{\text{EBITDA} - \text{Amortizacoes}}{\text{Total_activo}}$
- $\text{ROE (Return on Equity)} = \frac{\text{Result_Liq}}{\text{Capital_Proprio}}$

Missing values, zeros, and noisy values

The variable `Stock_turnover_per` is the only one to report missing values, with a total of 22 (8,8% missing values).

There are 9 variables that report zeros. The distribution of zeros can be seen in the following heatmap, where the zeros are coded in yellow:

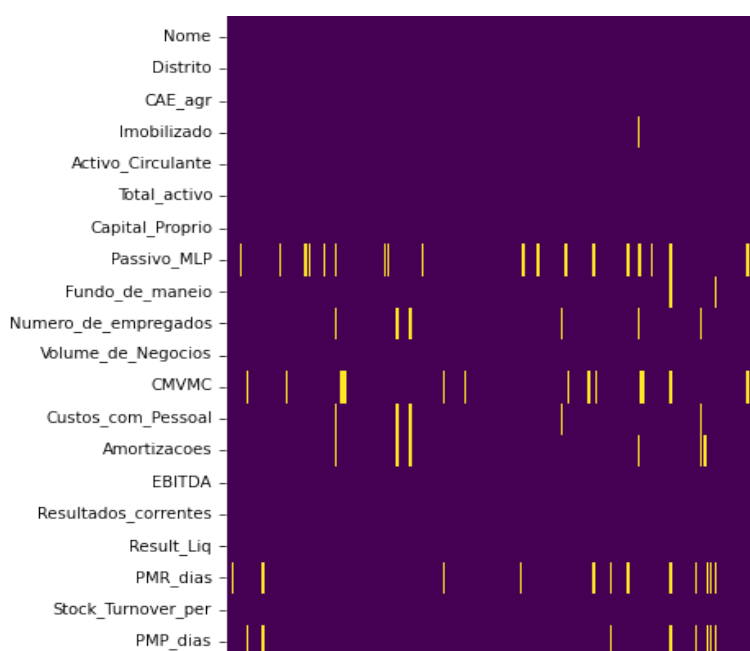


Figure 1 – Heatmap of Zeros across Observations

It's not possible to say for certain whether the zeros in each variable are missing values or actual zeros, we can only judge based on patterns across variables and on theoretical knowledge about the variables. From the graph presented above, we can observe that the zeros are concentrated in much the same observations, which may indicate that most of the zeros could be missing values. There are some cases where it's possible to deduce based on theory: zeros in "PMR_dias" (Average Collection Period) for retail companies should be actual zeros, zeros in "Amortizacoes" (Amortizations) for observations where "Imobilizado" (Fixed Assets) is non-zero must be missing values.

However, all variables have less than 9% of missing values plus zeros. There are also no values outside the expected range or without the expected format. Therefore, the dataset is overall well-composed and clean.

2 Univariate and Bivariate Analysis

2.1 Univariate Analysis

i. Numeric Continuous Variables

Our Dataset has 16 numeric continuous variables and one discrete variable. Given that the discrete variable has a considerable range (from 0 to 26.859) we will analyse it together with the continuous variables. Additionally, we will also analyse the 10 new ratio variables.

The first 6 variables are financial variables from the companies' Balance Sheet: Non-current assets, Current assets, Total assets, Total Equity, Non-current liabilities, and Working Capital. These variables present different scales, with the mean varying from € 20,9 million ("Fundo_de_Maneio") to € 682,5 million ("Total_do_ativo"). This difference in magnitude is expected, as some variables include the values of others: for example, "Total_do_ativo" includes both "Imobilizado" and "Ativo_Circulante".

	Imobilizado	Activo_Circulante	Total_ativo	Capital_Proprio	Passivo_MLP	Fundo_de_maneio
count	250	250	250	250	250	250
mean	396.292	286.194	682.486	195.559	205.858	20.887
std	1.792.516	1.015.393	2.714.305	962.209	1.060.755	108.583
Coefficient of variation	452%	355%	398%	-	515%	-
min	0	156	161	-2.826.846	0	-779.323
max	21.777.874	13.511.633	29.922.362	11.448.166	13.794.688	789.222
25%	14.841	61.787	94.103	16.706	1.372	-1.230
median	61.277	104.599	189.418	52.040	9.511	17.465
75%	153.083	195.314	368.263	133.054	40.075	42.755
IQR = Q3-Q1	138.241	133.527	274.160	116.349	38.703	43.985
Pearson_skewness	0,56	0,54	0,54	0,45	0,56	0,09
Kurtosis	107	124	100	96	120	30
Moderate Outliers	14	7	15	8	6	10
Severe Outliers	26	17	22	21	38	13

Table 1 - Financial Variables

Taking the mean values, we can describe the Dataset the following way: the "mean company" in the dataset has € 682,5 million in Total Assets, most of them allocated to Non-current Assets (€ 396 million). The company's assets are mostly financed by borrowed capital, as the Debt Ratio is higher than 70% (Financial Autonomy less than 30%). The mean working capital is around € 20,9 million.

Looking at dispersion measures, we find that all the variables present a significant range and that standard deviation presents relatively large values when compared to the mean, leading to a high Coefficient of Variation figures. The variable with the largest Coefficient of Variation is "Passivo_MLP", and the one with smallest (nevertheless, still a high value – 355%) is "Activo_Circulate".

Interquartile Range (IQR) values are more comparable between variables. Nevertheless, comparing the maximum value with the 3rd quartile for all variables, we find very different scales in values, which, again, is a marker for the presence of outliers. On that matter, the statistics for Moderate¹ and Severe² Outliers in the table above as well as the boxplots presented below confirm the presence of outliers (between 13 and 38 Severe Outliers per variable). There are eight companies that are severe outliers in all financial variables (except "Fundo_de_maneio"), that is, they are the largest companies by Balance Sheet:

¹ Considering D as the distance between the first and the third quartile ($D = Q3 - Q1$), an observation is considered Moderate Outlier if it lays between $Q1 - 3D$ and $Q1 - 1.5D$ or between $Q3 + 1.5D$ and $Q3 + 3D$.

² Considering $D = Q3 - Q1$, an observation is considered Severe Outlier if it is smaller than $Q1 - 3D$ or larger than $Q3 + 3D$.

Index	Nome	Total ativo
16	EDP - ENERGIAS DE PORTUGAL, S.A.	29.922.362
34	INFRAESTRUTURAS DE PORTUGAL, S.A.	28.874.983
25	EDP - GESTÃO DA PRODUÇÃO DE ENERGIA, S.A.	6.262.321
15	PETROGAL, S.A.	5.622.859
30	E-REDES - DISTRIBUIÇÃO DE ELETRICIDADE, S.A.	3.893.185
18	MODELO CONTINENTE - HIPERMERCADOS, S.A.	3.876.994
20	THE NAVIGATOR COMPANY, S.A.	3.435.313
44	SAIPEM (PORTUGAL) - COMÉRCIO MARÍTIMO, SOCIEDA...	2.741.896

Table 2 - Severe Outliers - Financial Variables

Form the table above we can conclude that are “financial outliers” mainly companies from the energy sector, but also public infrastructures, retail and paper.

Regarding the shape of the distribution of the variables, we found that that all the variables present significant differences between the mean and the median, with the mean being considerably higher than median, which, again, points us to the presence of outliers and/or *skewness* (positive *skewness*) in the distribution of the variables. In fact, the *Pearson's skewness degree*³ is positive in all variables: with the exception of “Fundo_de_maneyo”, all variables have *Pearson's skewness degree* around 0,5. Moreover, all the variables present relatively high values of kurtosis, that is, the distributions of the variables have “tails” heavier than those of the Normal distribution.

The following graphical representation of the variables will allow us to project a visual image of the description above.

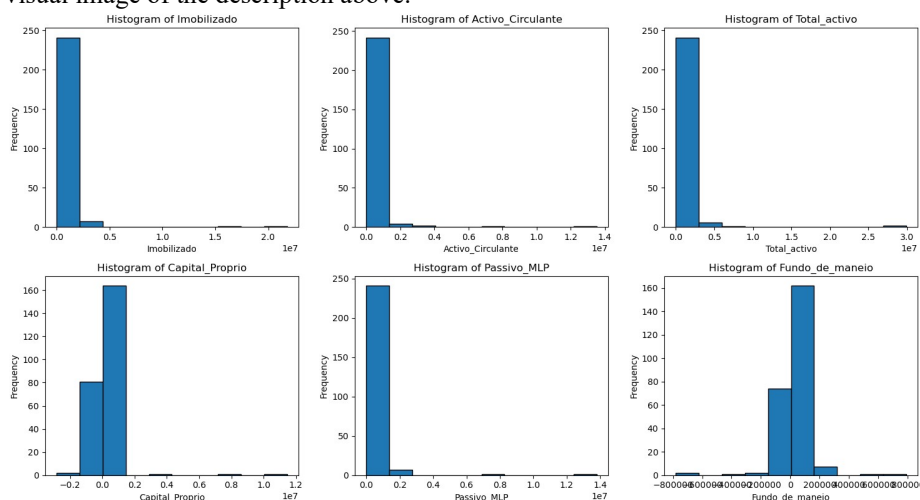


Figure 2 - Histograms for financial variables

³ $Pearson's\ Skewness\ degree = 3 \frac{\bar{X} - Md}{s}$, where $\bar{X}$ is the mean value, Md is the median value and s is the standard deviation. Considering the nature of the variables (continuous variables), our data has no significant mode. Given that, we considered this version of Pearson's Skewness degree (also known as Pearson's second coefficient of skewness) to be preferable, as it does not rely on mode as a measure of central tendency.

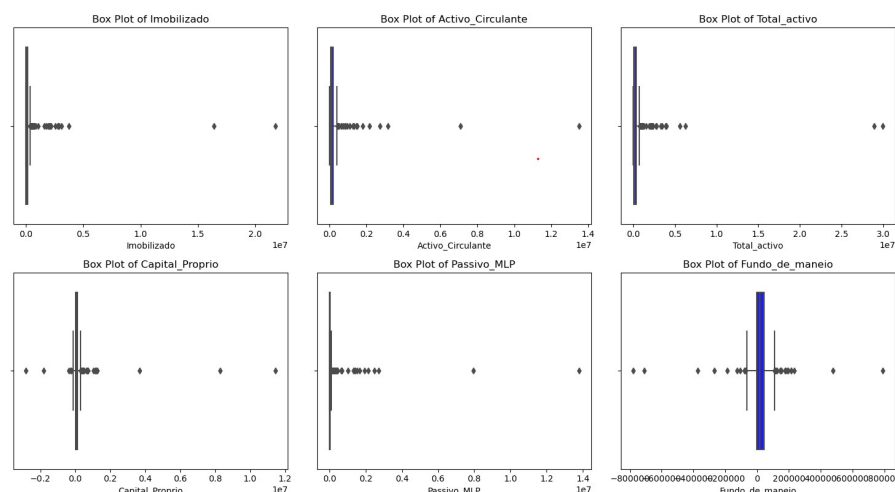


Figure 3 - Boxplots for financial variables

The next 7 variables are economic variables obtained from the company's Profit and Loss Statement (P&L): Turnover, Cost of goods sold and materials consumed, Employee benefits expenses, Depreciation and amortization expenses, EBITDA, Income from continuing operations before interests and tax and Net Income.

	Volume_de_Negocios	CMVCM	Custos_com_Pessoal	Amortizacoes	EBITDA	Resultados_correntes	Result_Liq
count	250	250	250	250	250	250	250
mean	609.104	453.136	44.766	22.168	44.954	24.750	18.420
std	1.280.090	1.181.408	70.592	52.184	97.739	99.271	82.984
Coefficient of variation	210%	261%	158%	235%	-	-	-
min	160.398	0	0	0	-98.121	-201.039	-158.947
max	14.864.570	13.493.390	487.455	395.612	582.888	941.682	848.565
25%	199.183	107.330	6.275	1.193	4.944	1.450	1.115
median	281.033	178.706	17.131	5.114	16.680	8.993	7.347
75%	476.325	326.883	48.068	15.043	46.143	23.098	19.075
IQR = Q3-Q1	277.142	219.552	41.793	13.850	41.200	21.648	17.961
Pearson_skewness	0,77	0,70	1,17	0,98	0,87	0,48	0,40
Kurtosis	70	70	12	20	12	45	53
Moderate Outliers	13	8	20	7	12	10	13
Severe Outliers	17	18	14	25	15	28	28

Table 3 - Economic Variables

Given the nature of the P&L, naturally the value of the variables decreases from left to right on the table, that is, as we go down the P&L. Nevertheless, these variables present a similar behaviour to that of the financial variables. The variables present different scales, as expected, given the nature of the variables: the mean varies from € 609,1 million in the variable “Volume_de_Negocios” to € 18,4 million in the variable “Result_Liq”.

Using the mean as reference for location measure, we can state that the “mean company” as a Turnover around € 609 million, with the Cost of Goods Sold and Materials Consumed being the main item of costs, corresponding to 74% of the turnover. The EBITDA reaches € 45 million (margin EBITDA of 7,4%) and the Net Income € 18,4 million (3,0% of the Turnover).

Dispersion wise, these variables also present significant range and the standard deviation displays relatively large values compared to the mean. Consequently, the Coefficient of Variation has relatively large figures, however they are in a smaller scale than the one observed in financial variables: “CMVCM” has the largest variation (Coefficient of Variation = 261%), and the smallest (158%) being “Custos_com_Pessoal”. On the other hand, the IQR interval is quite compact, which is an indicator for the presence of outliers: the statistics for Moderate and Severe Outliers in the table above confirm the presence of severe outliers (between 14 and 28

outliers per variable) and the box plots below give a visual representation of the outliers. There are no companies that are outliers in all of the economic variables, as companies from different sectors have quite different economic structure, meaning that a company may be an outlier in some economic variables but not on others. Still, we consider relevant to identify companies that are outliers in the main financial variables (Turnover and Costs of Goods Sold and Materials Consumed):

Index	Nome	Volume de Negocios	CMVMC
15	PETROGAL, S.A.	14.864.570	13.493.390
16	EDP - ENERGIAS DE PORTUGAL, S.A.	9.084.227	8.766.247
18	MODELO CONTINENTE - HIPERMERCADOS, S.A.	5.045.291	3.734.655
17	PINGO DOCE - DISTRIBUIÇÃO ALIMENTAR, S.A.	4.929.683	3.924.926
19	GALP - GÁS NATURAL, S.A.	4.509.051	4.483.789
21	SU ELETRICIDADE, S.A.	3.721.487	3.683.345
22	VOLKSWAGEN AUTOEUROPA, LDA	3.674.083	3.198.268
23	EDP COMERCIAL - COMERCIALIZAÇÃO DE ENERGIA, S.A.	3.559.012	3.127.844
20	THE NAVIGATOR COMPANY, S.A.	3.531.206	3.138.647
24	REPSOL PORTUGUESA, LDA	2.781.713	2.549.477
25	EDP - GESTÃO DA PRODUÇÃO DE ENERGIA, S.A.	2.374.308	2.106.287
26	ENDESA ENERGIA, S.A. - SUCURSAL EM PORTUGAL	2.053.683	1.961.152
27	BP PORTUGAL - COMÉRCIO DE COMBUSTÍVEIS E LUBRI...	1.913.003	1.821.226
28	EDP GEM PORTUGAL, S.A.	1.837.700	1.922.630
32	PRIO SUPPLY, S.A.	1.331.170	1.299.117

Table 4 - Severe Outliers - Economic Variables

As in the financial variables, the energy sector is the main sector in “economic outliers”. Also, there are 5 companies that are both “financial and economic outliers”: “Petrogal, S.A.”, “Edp - Energias De Portugal, S.A.”, “Edp - Gestão Da Produção De Energia, S.A.”, “Modelo Continente - Hipermercados, S.A.” and “The Navigator Company, S.A.”.

Looking at the shape of the distribution of the variables, as was the case with the financial variables, the economic variables present significant differences between the mean and the median, with the mean being considerably higher than median, which points us to the possibility of outliers and/or *skewness* (positive *skewness*) in the distribution of the variables. In fact, *Pearson's skewness degree*⁴ is positive in all variables: all variables have *Pearson's skewness degree* higher than 0,4, that is, these variables are relatively more skewed compared to the financial variables. The economic variables also present “tails” heavier than those of the Normal distribution, however in a lesser extent when compared with the financial variables.

Next, we present the graphical representation of the variables to provide a visual support to the description above.

⁴ Pearson's second coefficient of skewness

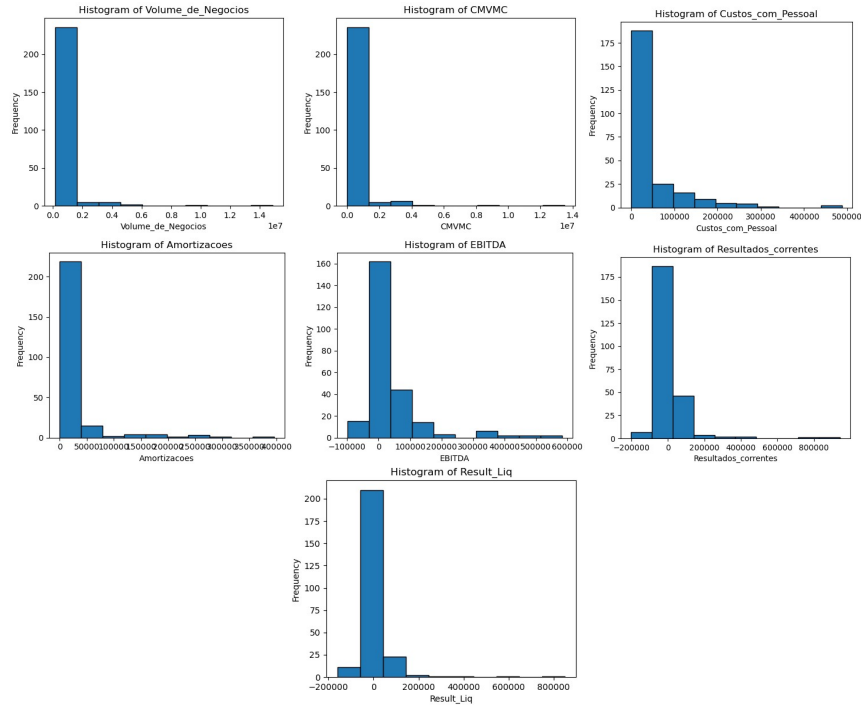


Figure 4 - Histograms of Economic Variables

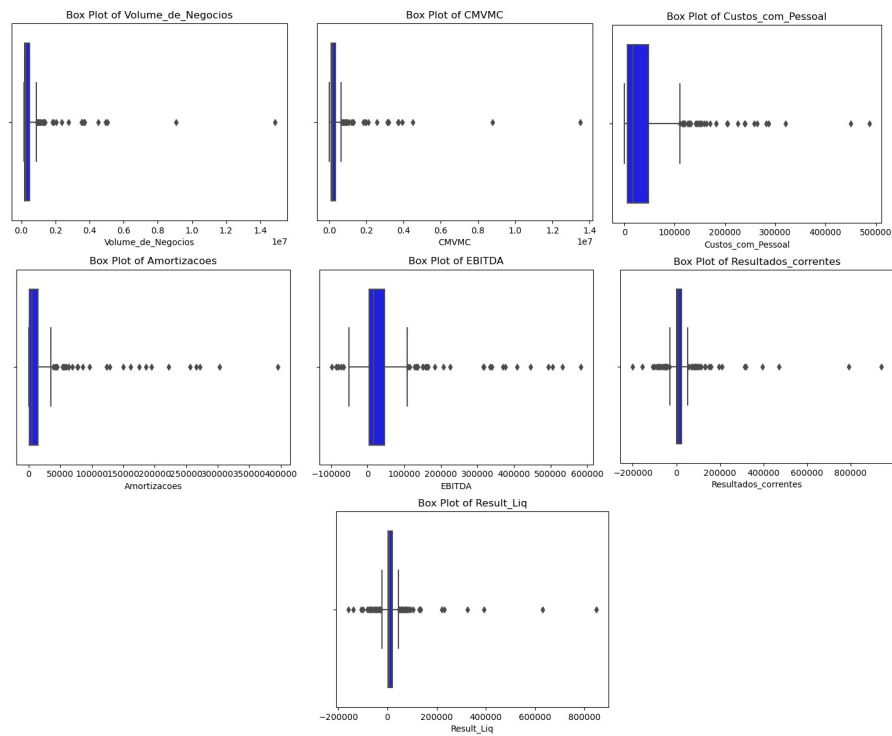


Figure 5 - Boxplots of economic variables

The last 3 original numerical continuous variables and one discrete variable are activity variables: Average collection period (days), Stock Turnover %, Average payment period (days) and Number of Employees. This last variable, Number of Employees is a numeric discrete variable, however, given its range (from 0 to 26.859), we will analyse it together with the continuous variables.

	Numero_de_ empleados	PMR_dias	Stock_Turnov er_per	PMP_dias
count	250	250	228	250
mean	1.447	43	1.564	47
std	3.016	41	22.484	35
Coefficient of variation	208%	97%	1437%	74%
min	0	0	1	0
max	26.859	230	339.542	210
25%	148	14	9	22
median	437	31	15	41
75%	1.396	61	33	63
IQR = Q3-Q1	1.248	47	25	40
Pearson_skewness	1,01	0,83	0,21	0,52
Kurtosis	39	4	228	3
Moderate Outliers	18	8	6	7
Severe Outliers	14	3	21	2

Table 5 - Activity Variables

Using again the “mean company” as reference, the Average collection period and the Average payment period are very close to each other, around 45 days. The stock Turnover is 1.564%, indicating that the “mean company” operates with small inventory when compared to the cost of goods sold, and the number of employees is close to 1.500.

The variables “Numero_de_empleados” and “Stock_Turnover_per” show a behaviour more in line with the financial and economic variables: very different values for the mean and median, high dispersion (large Coefficients of Variation, with “Stock_Turnover_per” standing out), positive *skewness* and “tails” heavier than those of the Normal distribution.

The variables PMR and PMP show a different behaviour to that of the other variables. These variables have similar magnitudes and reveal a mean and median relatively close to each other. They show comparatively less variation than the other variables, nevertheless, there are positively *skewed* and with “tails” heavier than those of the Normal distribution, as all the other variables.

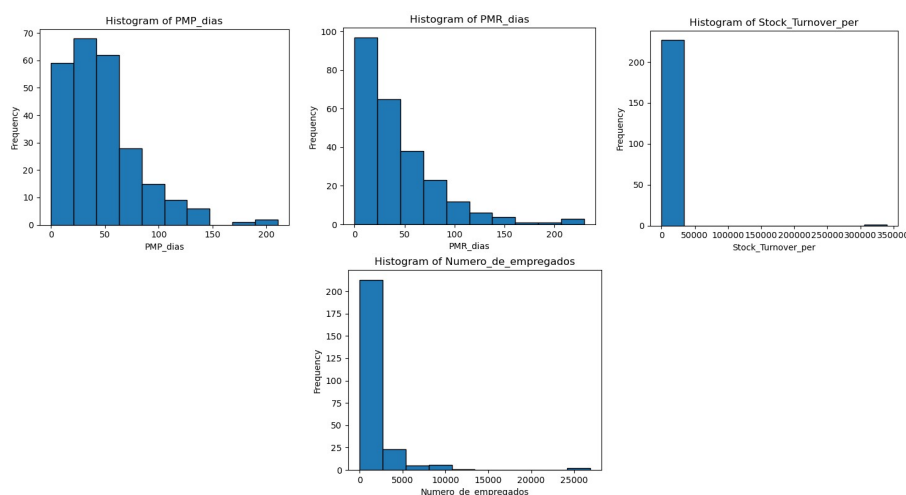


Figure 6 - Histograms of the Activity Variables

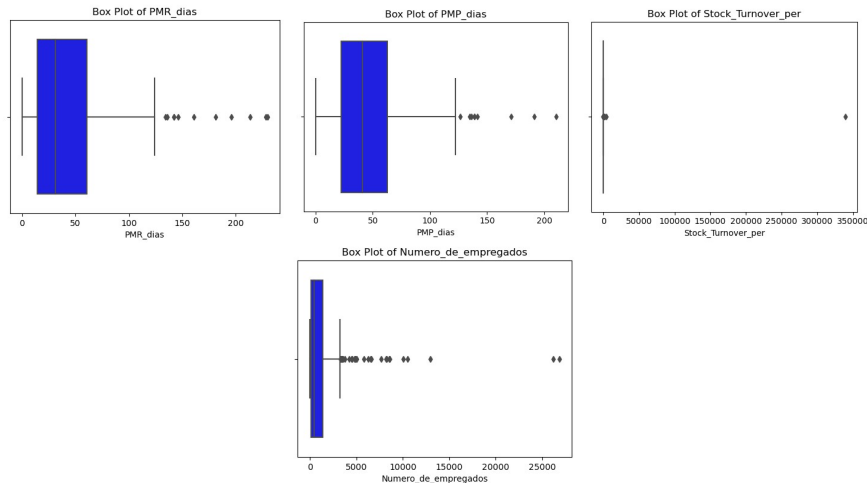


Figure 7 - Boxplots of the Activity Variables

The histograms of all numerical variables confirm that all variables are positively *skewed* and with “tails” heavier than those of the Normal distribution. The boxplots reinforce the *skewness* of the variables, showing also the presence of outliers in all variables (mostly upper outliers), however there are no companies that are outliers in all variables.

This Dataset is composed of data of the 250 largest companies in Portugal, by Turnover. The *skewness* and outliers identified in our analysis indicate that, among the 250 largest companies, there is a small group that we’ve identified as “the largest of the largest”: around 5% of the dataset are companies with a turnover between € 2 billion and € 15 billion. The other group (“the smallest of the largest”) are around 95% of the dataset and have a turnover between € 160 million and € 2 billion.

To end the analysis of the original Numeric Continuous Variables, we compare the distribution of the given variables with the normal distribution. To that end, the Q-Q plot will give us a clear understanding of the similarity between distributions.

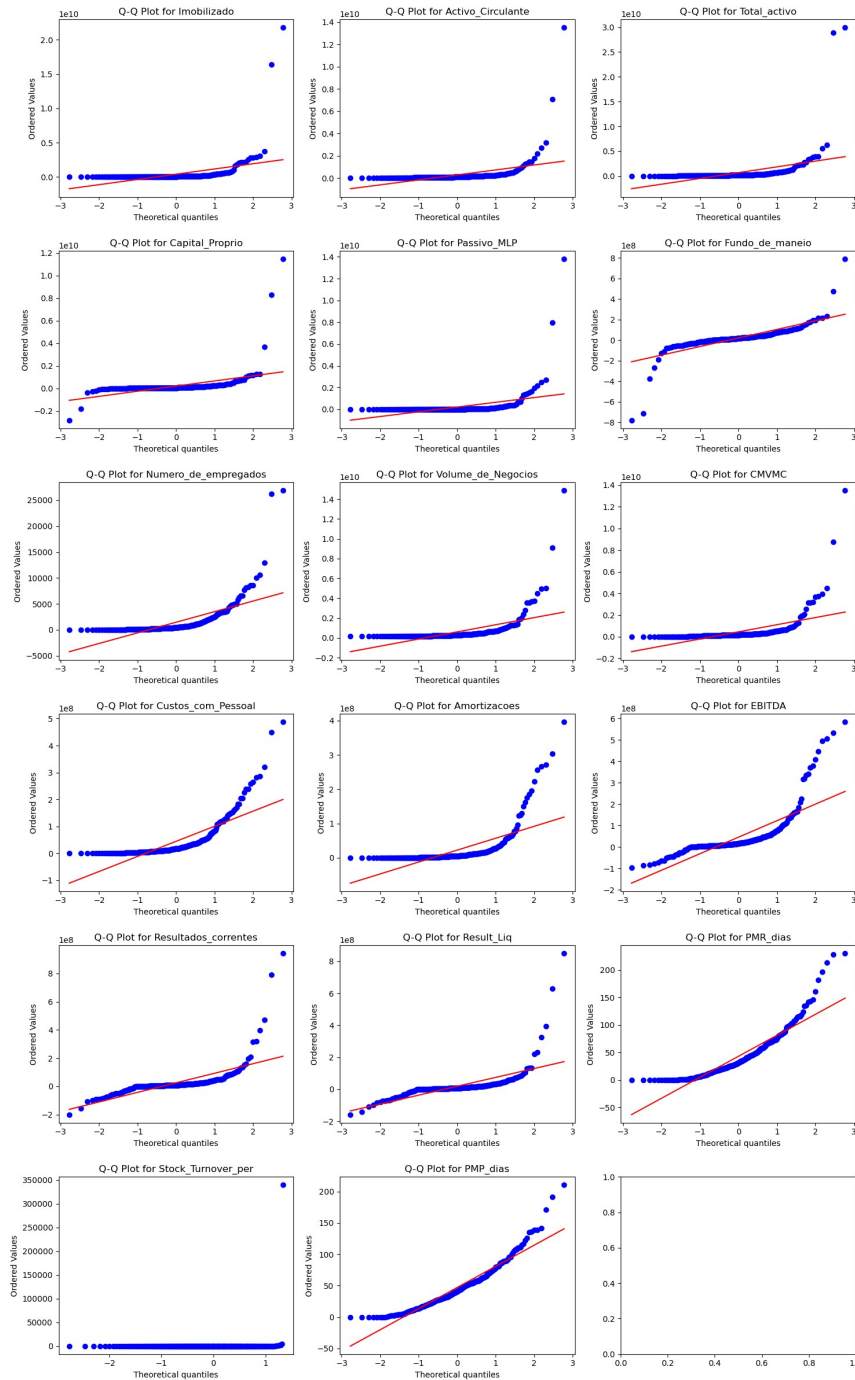


Figure 8 - Q-Q Plots

The plots show that a group of variables (“Imobilizado”, “Ativo_Circulate”, “Total_do_ativo”, “Capital_Proprio” “Fundo_de_manejo” and “PMP_dias”) that have a distribution closer to the Normal distribution, but not quite the same: the observed points are only near a straight line and with some deviations. Additionally, we can also see that these variables also exhibit severe outliers which deviates even further from the Normal distribution. The other variables are even further from the Normal distribution.

We also performed the Kolmogorov-Smirnov test (results of the test appended), and the null hypothesis (“The variable follows a normal distribution.”) was rejected for all variables, which leads us to conclude that our variables do not follow a normal distribution.

As stated in chapter 1, besides the original variables we introduced 10 new ratio variables.

	Racio_FM	Liquidez	Aut_Finac	Racio_Endiv	Solvabilidade	Debt_EBITDA	M_EBITDA	Rent_Vnd	ROI	ROE
count	249	250	250	250	250	250	250	250	250	250
mean	4.97	1.35	0.25	0.75	0.79	26.76	0.09	0.05	0.88	0.06
std	32.38	0.84	0.74	0.74	1.12	222.49	0.17	0.12	12.90	1.92
Coefficient of Variation	-	0.624	-	0.984	-	-	-	-	-	-
min	-282.12	0.06	-9.44	0.11	-0.90	-164.28	-0.35	-0.41	-0.54	-28.32
25%	0.89	0.88	0.17	0.52	0.21	1.91	0.02	0.01	0.02	0.07
median	1.24	1.15	0.31	0.69	0.45	5.59	0.05	0.03	0.06	0.17
75%	2.13	1.62	0.48	0.83	0.93	10.34	0.12	0.08	0.11	0.31
IQR = Q3-Q1	1.24	0.74	0.31	0.31	0.72	8.43	0.11	0.08	0.10	0.25
max	275.84	6.58	0.89	10.44	7.92	3397.26	1.02	0.59	203.95	4.86
Pearson_skewness	0.35	0.70	-0.24	0.24	0.90	0.29	0.74	0.44	0.19	-0.18
Kurtosis	58.31	7.97	123.72	123.72	12.41	213.97	7.96	4.85	249.94	192.04
Moderate Outliers	8	11	0	4	14	10	14	8	9	12
Severe Outliers	30	5	6	6	9	15	11	13	13	16

Table 6 - Ratio Variables

As expected, ratio variables present smaller scale than original variables. Still, comparing these variables with each other we conclude that they have different scales, with the mean value ranging from 26.76 of Debt_EBITDA to 0.06 of ROE.

Using these variables to characterize the “mean” company, we can state that the mean company has reasonable values of liquidity (1.35) and an equilibrated financial structure: non-current assets are financed only by permanent capital and current assets are sufficient to face short term financial demands. Financial Autonomy is around 25%, which is a relatively small value (relatively low levels of equity), and Debt to EBITDA is quite high (26.76). Margin EBITDA is close to 10%, the Return on Investment is quite high (0.88) contrary to the Return on Equity, which is quite low (0.06).

Dispersion wise, some variables, such as Racio_FM and Debt_EBITDA, present a large range of values and standard deviation relatively large when compared with the mean. However, most of the variables have a relatively smaller range and more comparable values between mean and standard deviation. As most variables don’t always have the same sign, Coefficient of variation is only available for two variables. The IQR is relatively contained across variables, which is an indicator of the presence of outliers, a fact that is confirmed on the bottom of the Table 6 by the Outliers rows. There are no companies that are outliers in all financial ratios and only two companies are outliers in all economic ratios: “Costaterra - Sociedade Imobiliária de Grândola.” and “Centro Hospitalar de Lisboa Ocidental, E.P.E”.

Looking at the shape of the distribution of the variables, contrary to the original variables, differences between the mean and the median are relatively contained for most variables, leading to values in *Pearson’s skewness degree* (Pearson’s second coefficient of skewness) relatively small, with two variables exhibiting negative skewness. These variables also present “tails” heavier than those of the Normal distribution. Graphical representation below confirms and completes this description.

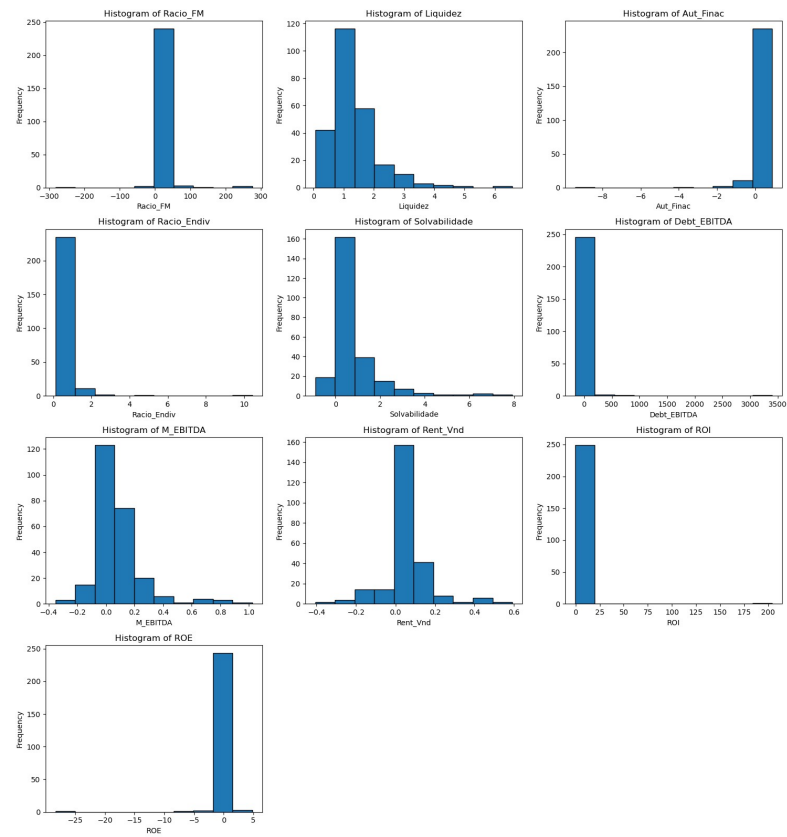


Figure 9 - Histograms Ratio Variables

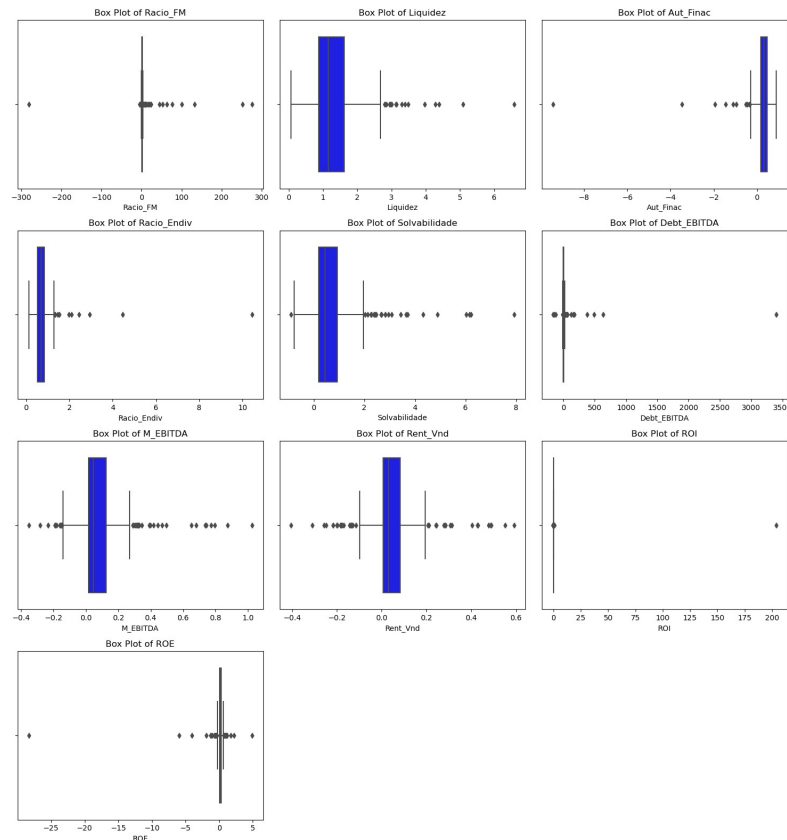


Figure 10 - Boxplots for Ratio Variables

The histograms and boxplots confirm that most variables are positively *skewed* (only two are negatively skewed), showing also the presence of outliers in all variables, some of them quite severe:

- *American Airlines, Inc*, for Rácio_FM, Auto_Fin, Racio_Endiv and ROI;
- *Rações Valouro, S.A.*, for Liquidez;
- *Agros* – for Solvabilidade;
- *Manuel Rui Azinhais Nabeiro, Unipessoal, Lda*, for Debt_EBITDA;
- *Centro Hospitalar Universitário Do Algarve, E.P.E.*, for ROE.

From the above it is clear that *American Airlines, Inc* is a company that is a severe outlier in several variables, a fact that should be noted when doing posterior analysis.

To end our univariate analyses, we compared the distribution of the given variables with the normal distribution, using the Q-Q plot.

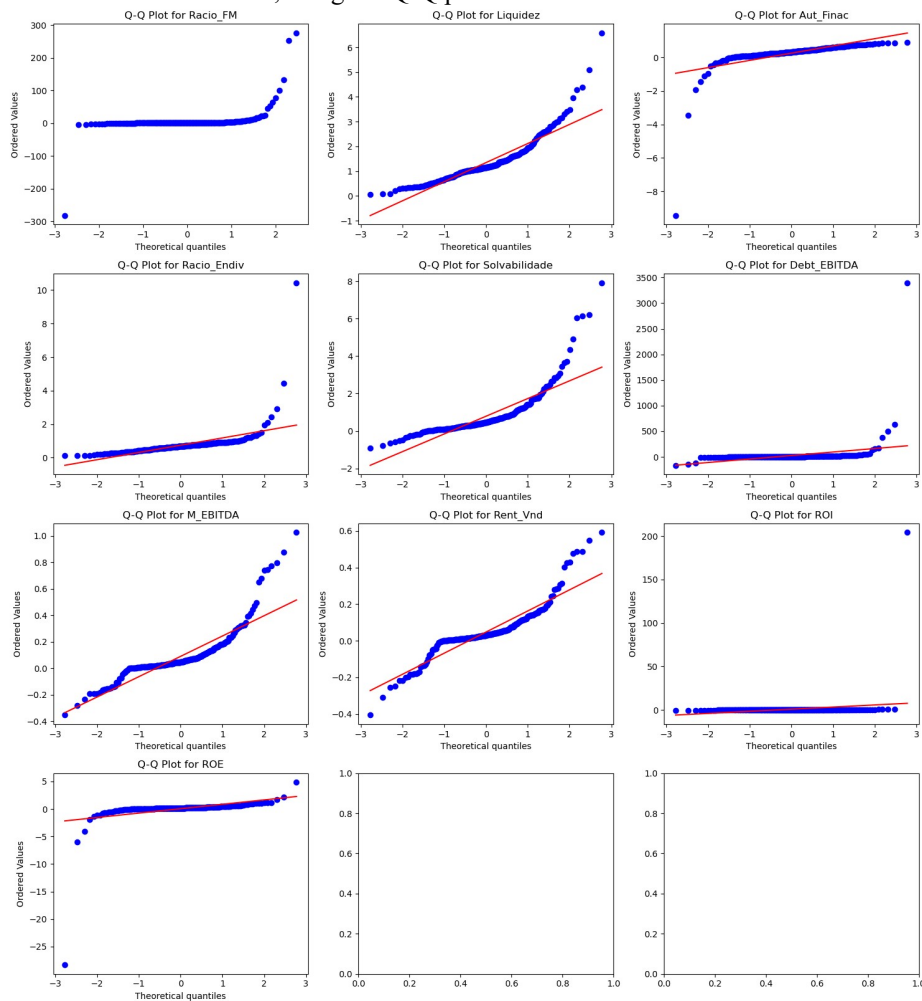


Figure 11 - Q-Q Plot for Ratio Variables

As it was the case with the economic and financial variables, ratio variables also don't seem to follow a normal distribution, a statement confirmed by the Kolmogorov-Smirnov test.

ii. Categorical Nominal Variables

Besides the 17 numerical variables, the Dataset has only 2 Categorical Nominal Variables: Distrito and CAE_agr, which are location (District) and activity variables (Activity Code).

Let's start by looking at the Bar diagrams for the absolute and relative frequencies.

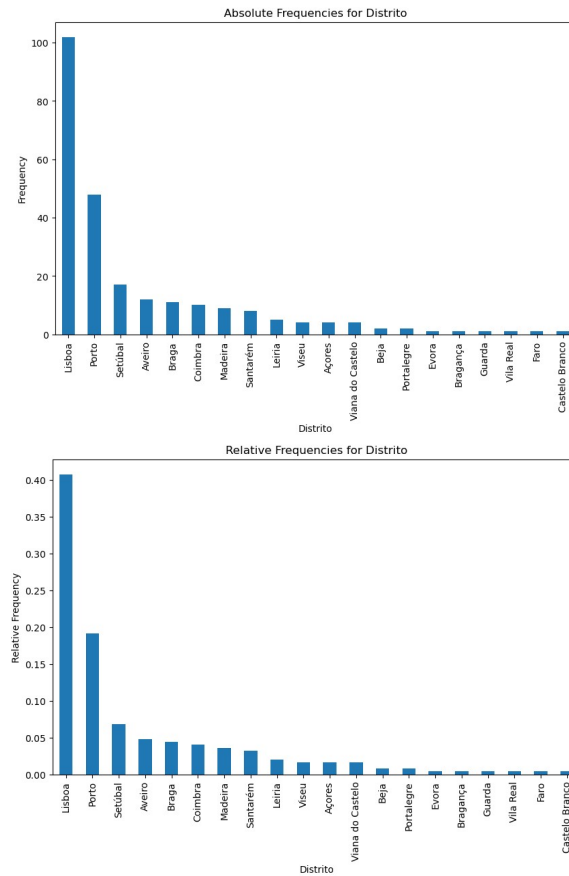
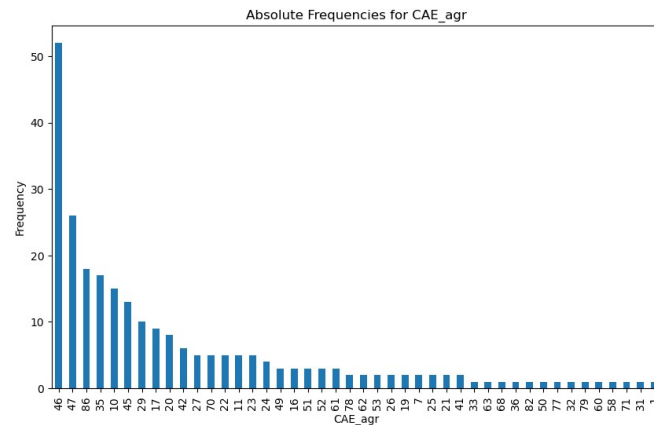


Figure 12 - Absolute and Relative Frequencies for Distrito

As expected, the region of Lisbon (Lisboa e Setubal) and the region of Oporto (Porto, Aveiro e Braga) is where more than 70% of companies are located, with the District of Lisbon being the mode with around 40% of the observations. Even though the largest companies concentrate their activity in the main two regions in Portugal, it's important to notice that all Districts in Portugal (the Autonomous Regions of Madeira and Azores formally are not Districts but were considered as such for the purpose of this study) have companies in the List of the 250 largest by Turnover.



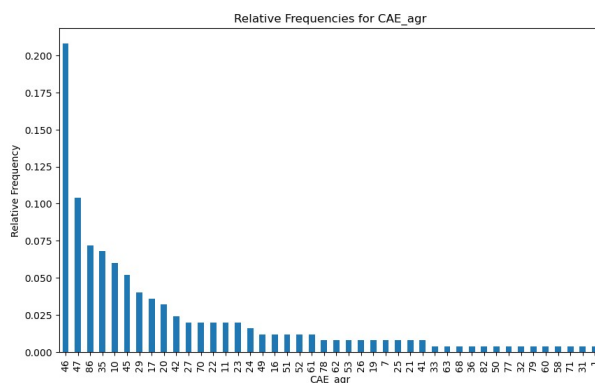


Figure 13 - Absolute and Relative Frequencies for CAE_agr

Activity wise, commerce is the most frequent activity in our database with more than 30% of observations: 46 – Wholesale, represents more 20% of the observations (around 50 companies with this activity) and 47 – Retail about 10% (around 25 observations).

The Health sector, the Energy sector, Food industry, Car sales and repair, are the next 4 most frequent, respectively. Combined, these activities represent more than 50% of our dataset.

The dataset comprises 44 activity sectors that include Agriculture, Extractive Industries, Manufacturing Industry, Utilities, Construction, Sales, Transport, Tourism, Communication, Consulting and Health: almost every sector of activity is represented in the 250 largest companies in Portugal and all major sectors of Economic Activity are represented: Primary Sector, Secondary Sector, Tertiary Sector and Quaternary Sector.

2.2 Bivariate Analysis

i. Pairs of categorical variables

In our Dataset we have only two categorical variables: Distrito and CAE_agr. As stated in the Univariate analysis, both variables have several categories, most of them with low frequency, a fact that is not favourable to the analysis of pairs of categorical variables. Having said that, we grouped the classes with lower frequencies (less than 10 observations) in both variables.

To analyse the relation between these two variables we created the following Contingency Table.

Distrito\CAE	10	35	45	46	47	86	Other	Total
Aveiro	1	0	0	1	0	0	10	12
Braga	0	0	1	1	1	1	7	11
Lisboa	7	10	7	25	13	7	33	102
Porto	2	3	4	9	7	5	18	48
Setúbal	1	0	0	2	1	1	12	17
Other	4	4	1	14	4	4	29	60
Total	15	17	13	52	26	18	109	250

Table 7 - Contingency Table

In the Univariate analysis of the Categorical variables, we have seen that 46 – Wholesale and Lisbon were the mode in the variables CAE_agr and Distrito, respectively, so it's unsurprising to find out that Wholesale activity in Lisbon has the highest frequency in the Contingency Table. In this table is possible to see that the second activity in the District with more companies (Lisbon) is 47 – Retail, followed by the Energy sector (this sector ranks 4th overall). The second District with more companies (Oporto) has also Wholesale, Retail and the Health sector as main

activities, mimicking the overall ranking. The distribution of companies by other Districts and Activities is much more disperse, with low frequencies in the pairs of categories.

To further study the relationship between these two variables, we tested the independence between them, performing the Chi-square test for Independence. Considering that this test is only valid when the expected frequencies are not too low (not more than 20% of the cross classes with expected frequencies less than 5 and none less than 1), we increased the threshold for the grouping classes to 17, for the purpose of this test.

The results of the test were the following:

Chi-square statistic: 7.1731356198267955
P-value: 0.3051322197090517
Degrees of freedom: 6

Considering that P-values is quite high (about 30,5%), we don't reject the null Hypothesis, that is, we don't to reject Ho: "There is no relationship between the Location and the Activity of companies".

Even though we did not conclude that the variables are dependent, we computed the Association Measures based on the Chi-2 statistics:

Phi coefficient: 0.14283944639954407
Pearson's C coefficient: 0.3301545781905634
Cramer's V coefficient: 0.15642060750206696

As expected, considering the result of the Chi-square test for Independence, all the association measures reveal low levels of association between the variables Distrito and CAE_agr.

ii. Pairs of numerical variables

Our Dataset has 17 numeric variables and, in this section, we are going to analyse how they relate with each other. As stated in the Univariate Analysis, we have 3 groups of numerical variables: Financial Variables, Economic Variables and Activity Variables.

Beforehand, given the nature of the variables, it's expected these variables to be related among them in each group and also, to a less extent, to be related across different groups.

To assess the relation between the pairs of numerical variables, we start of by looking at the graphical representation through a scatter plot, to get a first look and feel of the relation between the variables. A matrix of scatter plots for all pairs of variables is available in the appendix. From the matrix it's possible to identify the existence of correlation between variables, some more than others. For instance, there's a clear positive linear correlation between "RL" and "Resultados_correntes", between "Volume_de_Negócios" and "CMVMC". Other pairs of variables also reveal positive correlation, however the power of graphical representation is somehow lost due to the existence of outliers, as is the case of "Imobilizado" and "Ativo_circulante", "Imobilizado" and "Total_do_ativo". Lastly, there are pairs of variables where no pattern of correlation is visible, as is the case of "Fundo_de_maneyo" and "EBITDA", or "PMP" and "EBITDA".

Given the above, we are led to believe that there is correlation between variables. It is also clear that variables of the same group are more prone to be correlated with each other rather than to be correlated with variables from different groups.

To have a better understanding of how the variables relate, we calculated the correlation coefficient between the variables. Given that in the Univariate analysis we

concluded that our variables do not follow a normal distribution, we calculated the correlation using the Spearman's Ordinal Correlation Coefficient.

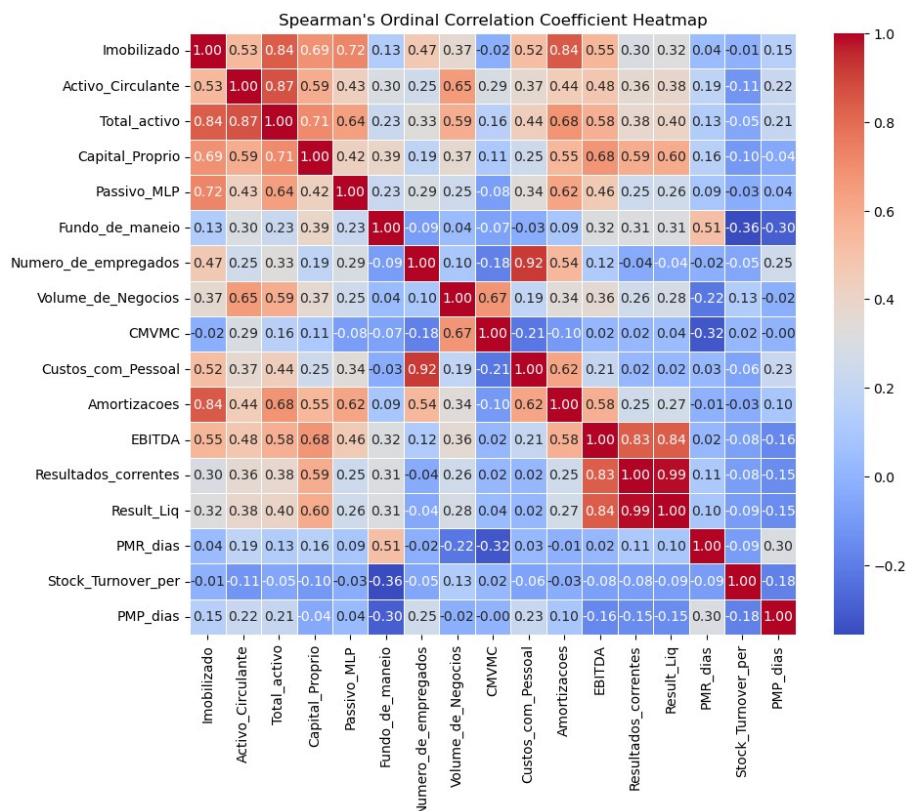


Figure 14 - Correlation Heatmap

The results of the Spearman's Ordinal Correlation Coefficient show us that the Activity Variables have no significant correlation (correlation coefficient higher than 0,75 or less than -0,75) with other variables (either from the same group or from other groups), except for one case: (Custos_com_Pessoal, Numero_de_empregados).

On the other hand, there is significant correlation between some pairs of Economic Variables: (Result_Liq, EBITDA), (Result_Liq, Resultados_correntes) and (EBITDA, Resultados_correntes). There is also a significant correlation between one Economic Variable and one Financial Variable: (Amortizacoes, Imobilizado).

Finally, looking at the Financial Variables we also find strong correlation between variables: (Total_do_ativo, Imobilizado) and (Total_do_ativo, Ativo_circulante). There are also pairs of variables with a correlation near to 0,75, namely (Capital_Proprio, Total_do_ativo) and (Passivo_MLP, Imobilizado).

Looking at the pairs of variables with strong correlation, we can conclude that the correlation is related to the known accountable nature of the variables:

- Custos_com_Pessoal, Numero_de_empregados: 0,92 – This a natural correlation: Number of employees and Employee benefits expenses. As the number of employees increases the Employee benefits expenses should increase in a very similar proportion;
- Result_Liq, EBITDA: 0,84. Net income is the EBTDA deducted of amortizations, financial cost and taxes, so it is expected to find a strong positive correlation between the two;
- Result_Liq, Resultados_correntes: 0,99. Net income is the Current Results deducted of Financial Cost and Taxes. This, way, these are relatively similar variables, being no surprise the existence of a very strong correlation;

- EBITDA, Resultados_correntes: 0,83. As Current Results is the EBITDA deducted of Amortizations and Financial Results, these are closely related variables;
- Amortizacoes, Imobilizado: 0,84. Amortisations registers the depreciation in value of Non-Current Assets. Thus, Amortisations depends directly of Non-Current Assets, and a strong correlation is to be expected;
- Total_do_ativo, Imobilizado and Total_do_ativo, Ativo_circulante: 0,84 and 0,87, respectively. Total Assets is equal to the sum of Non-Current Assets and Current Assets. This way, the larger (smaller) the Current Assets or Non-Current Assets the larger (smaller) the Total Assets, which means that a strong correlation between the variables is to be expected;
- Capital_Proprio, Total_do_ativo: 0,71. Total Assets are equal to Equity and Liabilities. This correlation is not as strong as the one on Total Assets and Non-Current Assets or Total Assets and Current Assets because, unlike Current and Non-Current Assets, Equity may be negative (it has a larger range) and it is possible to have an increase in Total Assets without a corresponding increase in Equity (if the increase in Total Assets is financed by borrowed capital), so a weaker correlation is to be expected;
- Passivo_MLP, Imobilizado: 0,72. It's common to finance Non-Current Assets with Long Term Liabilities, and this correlation is somehow evidence of that current practice.

Finally, it should be noticed that all significant correlations are only positive correlations (there is no significant negative correlation between variables).

The following scatter plot illustrates the correlations listed above.

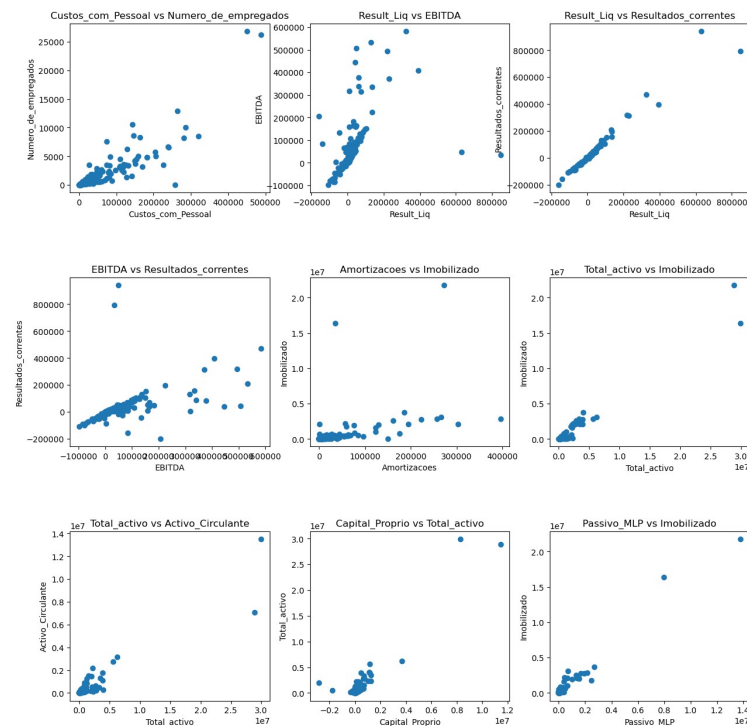


Figure 15 - Scatter Plot

It's noticeable the effect of outliers on the scatter plot, which leads to the loss of the representation power of the plot. To have a better image of the correlation between the paired variables, we've removed the most severe common outliers (list of the excluded outliers in appendix III).

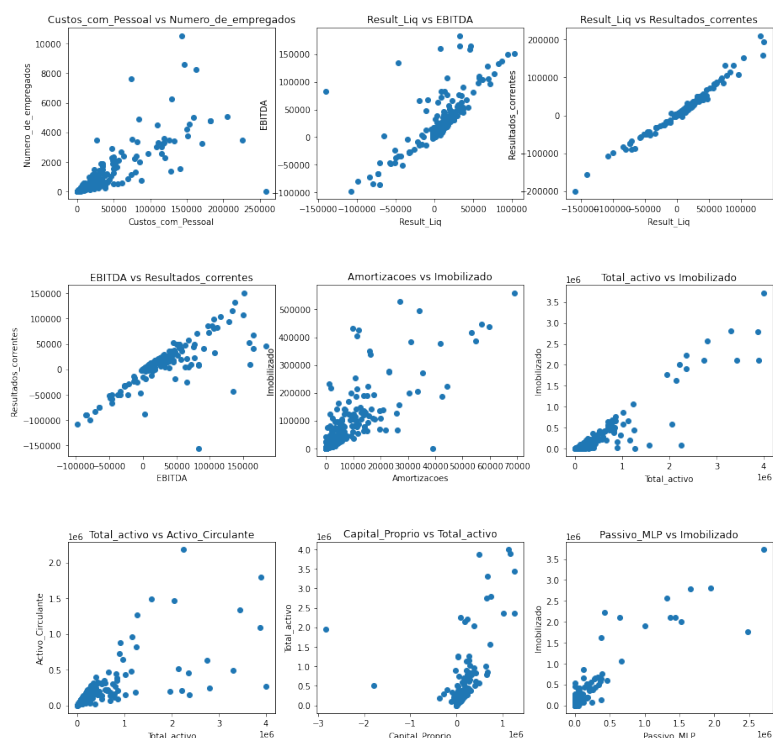


Figure 16 - Scatter Plot without most sever outliers

Even though the existence of some outliers is still evident (we opted to remove as little observations as possible so not to excessively disturb the representation of the data set) the pattern of correlation is now more evident in the scatter plot.

In conclusion and as expected, we've identified significant correlation between pairs of variables, most of them comprising variables form the same group and reflecting a known accounting relation.

iii. Pairs of numerical and categorical variables

Considering that most of the variables are numerical and only two variables are categorical, we will start by presenting the bivariate analysis of the numerical variables with the categorical variable “Distrito”, and then for “CAE_agr”. For this purpose, we will present boxplots of the numerical variables across the categorical variable, and then, to solidify our observations we will run a hypothesis test.

When analysing the distribution of the numerical variables by District, we observed that the category of Lisboa stands out: while the median is not particularly different from other categories, the amount of (mostly positive) outliers is generally higher for companies located in Lisboa, across the numerical variables. The same holds, to a lesser extent, for Porto and Setúbal. This can be seen, for instance, in the following boxplots (the boxplots for the remaining variables can be found in the appendix):

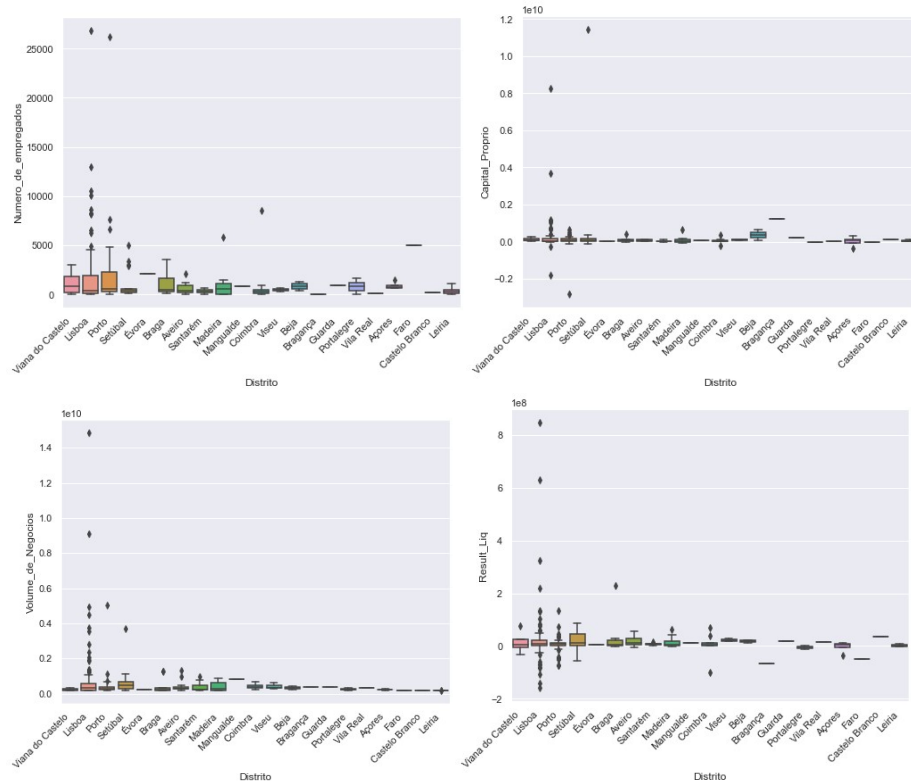


Figure 17 – Boxplots of the numerical variables by Distrito category

Since there are more than two Distrito categories and the variables don't follow a Normal distribution, we performed the Independent Samples Kruskal-Wallis' test on SPSS, for the hypotheses of same distribution across categories, and same median across categories, for each numerical variable. At a significance level of 5%, we only reject the hypothesis that "The medians of CMVMC are the same across categories of Distrito.". The SPSS output can be found in the appendix.

Thus, while the categories of Lisboa and to a lesser extent, Porto and Setúbal stand out graphically, the differences are not statistically significant in most variables.

When we analyse the distribution of the numerical variables by CAE_agr category, through the boxplots, we notice that there are some categories that stand out in accordance with our theoretical expectations about the economical/financial structure of companies.

As can be seen in the boxplots presented below, companies in category 52 (Storage and other auxiliary activities to transportation), represented by the blue boxplots, have a noticeably higher value in assets, and also in equity and medium/long-term debt. Companies in category 19 (Production of petroleum products), represented by the orange boxplots, have a noticeably higher Turnover and Cost of Goods Sold and Materials Consumed. Companies in categories 52, 19 and 61 (Telecommunications, purple boxplot) have the highest value of Amortizations. The remaining boxplots can be found in the appendix.

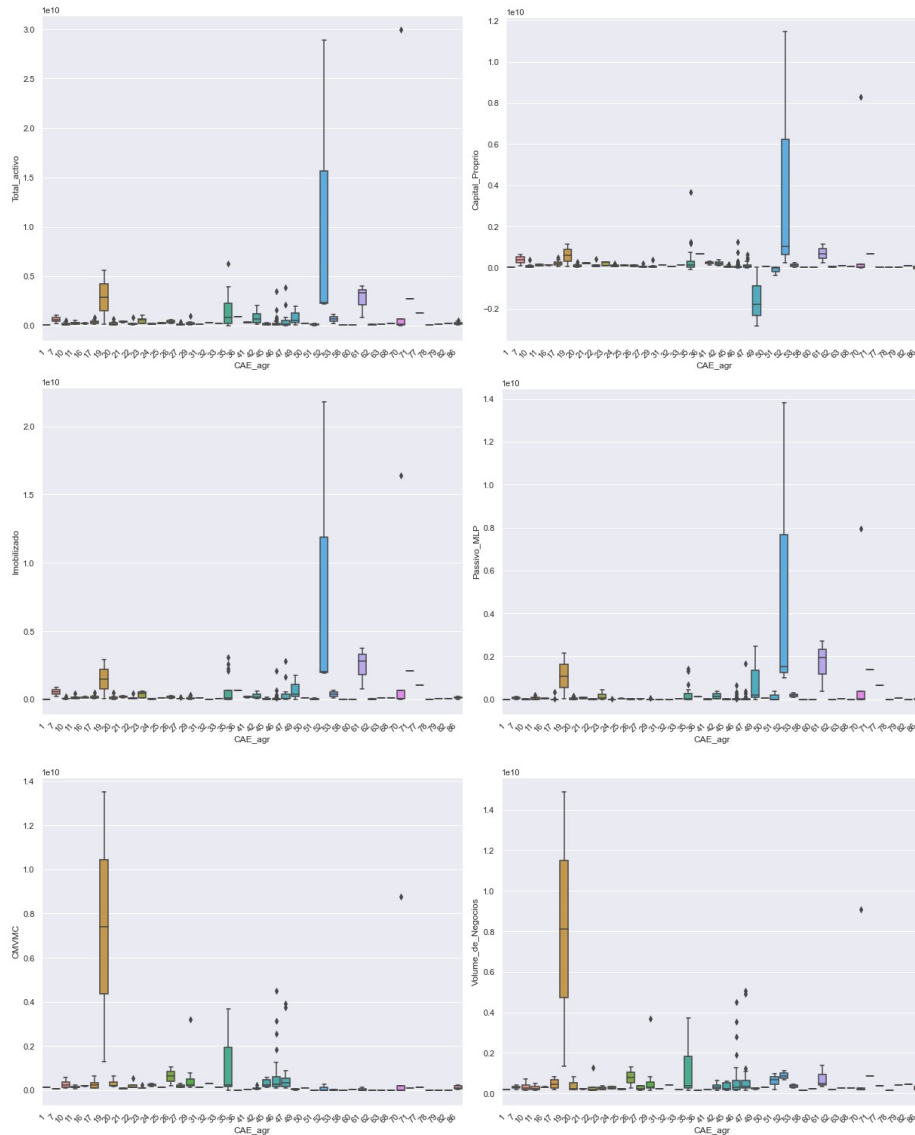


Figure 18 – Boxplots of the numerical variables by CAE_agr category

Since there are more than two CAE_agr categories and the variables don't follow a Normal distribution, we performed the Independent Samples Kruskal-Wallis' test on SPSS, for the hypotheses of same distribution across categories, and same median across categories, for each numerical variable. At a significance level of 5%, we reject almost all the null hypotheses. We only retain the hypothesis of same medians of Activo_Circulante, and same medians and distributions of Volume_de_Negocios across categories. The SPSS output can be found in the appendix.

Thus, CAE_agr appears to be a particularly important variable in this dataset, with interesting relations with the numerical variables. As we know from theory, the type of economic activity of a company influences its economic and financial structure, which is measured by the numerical variables of this dataset.

iv. Pairs of Ratio Variables

Ratio variables will be used separately from other variables in Cluster Analysis, reason why we are interested to analyse how they relate with each other.

To have a better understanding of how the ratio variables relate, we calculated the correlation coefficient between the variables (Spearman's Ordinal Correlation Coefficient).

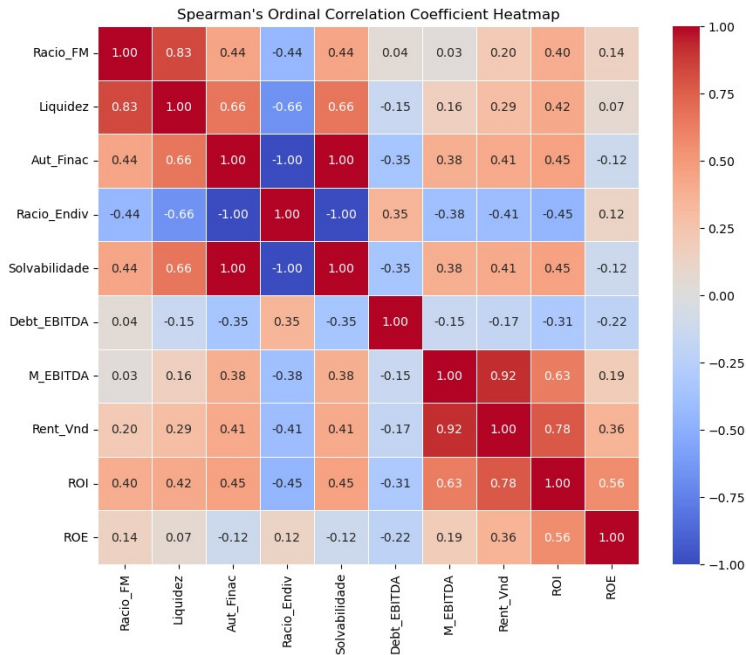


Figure 19 - Correlation Heatmap - Ratio Variables

The results of the Spearman's Ordinal Correlation Coefficient show us that there are a few pairs of variables with strong correlation:

- (Solvabilidate, Aut_Finac): 1. Whereas Solvability measures the relation between Equity and Liabilities, Financial Autonomy measures the relation between Equity and Total Assets (which are equal to Liabilities plus Equity). Even though these two variables take different views on the financial structure, they are closely related in their definition.
- (Solvabilidate, Racio_Endiv): 1. Racio_Endiv corresponds to the quotient between Liabilities and Total Assets, so, given the previous point, it's understandable the high correlation.
- Given the previous two correlations, it follows that (Racio_Endiv, Aut_Finac) are also strongly correlated;
- (Liquidez, Racio_FM): 0.83. These two ratios are complementary, which explains the high correlation. Whereas liquidity measures the relation between non-current assets and short-term liabilities, Racio_FM does the same for the long-term maturities on both sides of the balance sheet;
- (Rent_Vnd, M_EBITDA): 0.92. These are two profitability ratios that use the same variable from the P&L on the denominator (Turnover), which justify the high correlation.
- (ROI, Rent_Vnd): 0.76. These two ratios use the same variable on the numerator (Current Results), leading to a high correlation.

The scatter plot bellow (it excludes severe outliers identified in univariate analysis) illustrates the correlations listed above.

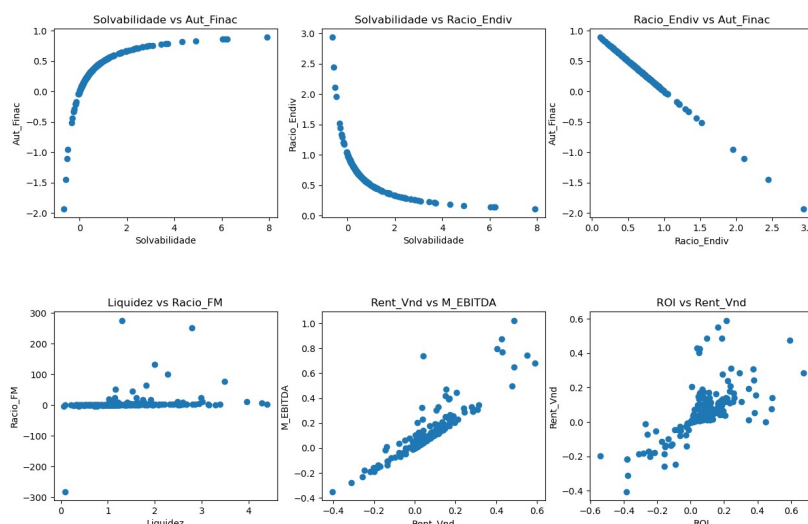


Figure 20 -Scatter Plot for Ratio Variables

As expected, there is some strong correlation between some pairs of variables, as they share some components in their definition or are complementary.

3 Multivariate Analysis

3.1 Principal Component Analysis

i. Principal Component Analysis of the Original Variables

Principal Component Analysis (PCA) is a data analysis technique that consists in transforming a set of correlated variables into a new set of variables that are uncorrelated but still comprise a large part of the original variable's information. These new variables are called Principal Components and are linear combinations of the original variables. This technique helps to reduce the dimension of the dataset, by transforming the correlated variables, while still retaining their information.

In the exploratory data analysis, it was possible to observe that a large part of the variables was substantially correlated with each other, which makes Principal Components Analysis interesting to implement. The following will be selected as Active variables for the PCA:

- *Imobilizado*
- *Activo Circulante*
- *Total_activo*
- *Capital_Proprio*
- *Passivo_MLP*
- *Fundo_de_maneio*
- *Numero_de_empregados*
- *Volume_de_Negocios*
- *CMVMC*
- *Custos_com_Pessoal*
- *Amortizacoes*
- *EBITDA*
- *Resultados_correntes*
- *Result_Liq*

The remaining 6 variables will be left as Supplementary variables, which means that they won't be used to compute the Principal Components but can still be projected into the principal component axes to be included in the analysis. These

Supplementary variables are *Nome*, *Distrito* and *CAE agr*, as they are categorical variables for which PCA is not effective; and *Stock Turnover Per*, *PMP_dias* and *PMR_dias*, as they didn't have as high levels of correlation as the other variables.

To implement the PCA, we have to consider whether our data is homogeneous or heterogeneous. All variables but one are in the same unit of measure (thousands of euros), but as seen in the exploratory data analysis, the variables have substantially different dispersions (their standard deviation ranges from 3.016 to 2.714.305). Thus, our data is heterogeneous. PCA doesn't work well with variables that have different scales, because the principal components may be biased towards the variables with larger magnitudes. Therefore, we apply Normed PCA, where the variables are standardized to solve this problem, and the correlation matrix is used instead of the covariance matrix.

Finally, we selected all cases (companies), with equal weights.

The Principal Component Analysis is then computed using the SPAD software, with the following parameters:

Method settings	
Type of analysis	Normed PCA
Weight variable for cases	None
Weighting of the variables	Uniform
Filter	No
Number of active cases	250
Number of supplementary cases	None
Number of active variables	14
Number of continuous supplementary variables	4
Number of categorical supplementary variables	2

Figure 21- PCA parameters in SPAD

We obtained the following axes and respective variance (eigenvalues):

Axis	Variance of the axis (eigenvalue)	% of explained variance	Cumulated % of explained variance
1	4,849	34,6	34,6
2	2,351	16,8	51,4
3	1,822	13,0	64,4
4	1,560	11,1	75,6
5	0,884	6,3	81,9
6	0,733	5,2	87,1
7	0,494	3,5	90,7
8	0,392	2,8	93,5
9	0,347	2,5	95,9
10	0,228	1,6	97,6
11	0,153	1,1	98,7
12	0,117	0,8	99,5
13	0,053	0,4	99,9
14	0,019	0,1	100,0
Total	14,000	100,0	100,0

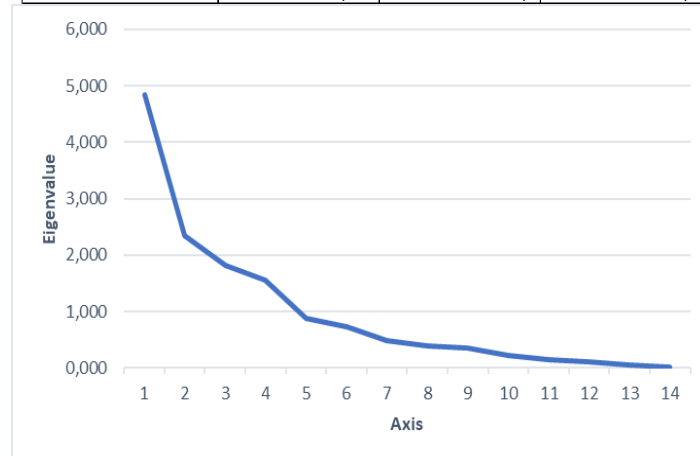


Figure 22- Eigenvalues

Considering these results, we can now decide how many axes we should keep in our analysis, so as to reduce the dimension of our data while still obtaining an informative set of variables. By Kaiser's criterion for selecting the number of axes to retain, we select the first 4 axes, as they have eigenvalues higher than 1. By Cattell's criterion, we may select the first 5 axes, as the 5th axis is the last one to have a relatively large difference in eigenvalue to the previous axis. By Pearson's criterion, we also select the first 5 axes, as they already explain about 80% of the variance.

Overall, considering all of the criteria, we decide to retain the first 5 axes.

In the following table, we present the coordinates of the original variables (both the active and the supplementary) in each axis. Since this is a Normed PCA, they also correspond to the correlations between the variables and the principal components:

Label of the variable	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
Imobilizado	-0,769	0,005	0,169	-0,053	-0,109
Activo_Circulante	-0,768	-0,054	-0,242	0,173	-0,287
Total_activo	-0,595	-0,244	-0,182	0,260	-0,522
Capital_Proprio	-0,664	0,264	0,058	-0,228	-0,255
Passivo_MLP	-0,771	-0,033	0,207	-0,318	0,271
Fundo_de_maneyo	0,027	0,536	0,074	-0,453	-0,457
Numero_de_empregados	-0,343	-0,810	0,241	0,233	-0,001
Volume_de_Negocios	-0,474	-0,134	-0,780	-0,110	0,151
CMVMC	-0,110	-0,134	-0,932	-0,081	0,107
Custos_com_Pessoal	-0,344	-0,769	0,265	0,160	-0,030
Amortizacoes	-0,709	-0,122	0,125	-0,526	0,195
EBITDA	-0,843	0,194	0,154	-0,230	0,216
Resultados_correntes	-0,564	0,543	0,044	0,574	0,140
Result_Liq	-0,523	0,547	0,045	0,587	0,175
CAE_agr	-0,042	-0,315	0,198	0,105	0,112
PMR_dias	0,019	0,122	0,117	-0,040	-0,318
Stock_Turnover_per	-0,222	0,118	0,111	-0,054	0,221
PMP_dias	0,016	-0,270	-0,095	0,163	-0,122

Figure 23- Coordinates of the original variables

We can see that most of the variables are negatively correlated with the principal components. The first principal component is most correlated with the *EBITDA*, and then also with the *Imobilizado*, *Activo_Circulante*, and *Passivo_MLP*. The second principal component is most correlated with *Numero_de_empregados* and *Custos_com_Pessoal*. The third principal component is most correlated with *Volume_de_Negocios* and *CMVMC*. The 4th and 5th principal components already have generally lower levels of correlation with the variables, but they are most correlated with *Amortizacoes*, *Resultados_correntes* and *Result_Liq*, in the case of the 4th component, and with *Total_Activo* and *Fundo_de_Maneio* in the case of the 5th.

Therefore, it is possible to find a "theme" to each component, particularly the first 3 components: the first relates to the profit and the company financial structure, in terms of assets and liabilities; the second relates to the employees and their costs; and the third relates to the sales and their costs. We will analyse this further when looking at each Principal Component.

Now, this table presents the eigenvectors of the 5 axes, for each of the original variables (active). The eigenvectors are the coefficients of the variables for each of the 5 principal components (which are linear combinations of these variables):

Label of the variable	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
Imobilizado	-0,349	0,003	0,125	-0,042	-0,116
Activo_Circulante	-0,349	-0,035	-0,179	0,138	-0,305
Total_activo	-0,270	-0,159	-0,135	0,208	-0,555
Capital_Proprio	-0,302	0,172	0,043	-0,182	-0,271
Passivo_MLP	-0,350	-0,022	0,154	-0,255	0,288
Fundo_de_maneyo	0,012	0,350	0,055	-0,363	-0,486
Numero_de_empregados	-0,156	-0,528	0,178	0,186	-0,001
Volume_de_Negocios	-0,215	-0,087	-0,578	-0,088	0,161
CMVMC	-0,050	-0,088	-0,691	-0,065	0,114
Custos_com_Pessoal	-0,156	-0,502	0,196	0,128	-0,032
Amortizacoes	-0,322	-0,080	0,093	-0,421	0,207
EBITDA	-0,383	0,126	0,114	-0,184	0,230
Resultados_correntes	-0,256	0,354	0,032	0,459	0,149
Result_Liq	-0,237	0,357	0,034	0,470	0,186

Figure 24- Eigenvectors

We can draw the very same conclusions from the eigenvectors that we drew from the coordinates. As expected, the variables with the highest coefficients in each axis are the ones that also had the highest values in coordinates. These findings are also reaffirmed by the contributions of each variable to the axes, which can be found in the appendix.

We can also analyze the principal components by cases. In the following tables, we present the coordinates, absolute contributions and relative contributions (square cosines) for the first and last 5 cases.

Active cases coordinates					
Identifier	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
case1	-0,676	0,978	0,323	0,842	0,004
case2	0,915	0,499	1,006	0,419	0,307
case3	-0,072	0,528	1,080	-1,469	1,245
case4	1,172	-0,613	1,179	0,317	0,145
case5	0,914	-0,418	0,477	0,428	0,310
...	...	...	...	...	...
case246	1,224	0,438	0,553	-0,017	-0,069
case247	0,527	0,735	0,743	-0,273	-1,154
case248	-2,011	-1,920	-1,468	-0,374	-0,507
case249	0,974	0,155	0,672	0,243	0,232
case250	-0,625	-0,805	-0,440	0,345	-0,082
Active cases contributions (%)					
Identifier	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
case1	0,0	0,2	0,0	0,2	0,0
case2	0,1	0,0	0,2	0,0	0,0
case3	0,0	0,0	0,3	0,6	0,7
case4	0,1	0,1	0,3	0,0	0,0
case5	0,1	0,0	0,0	0,0	0,0
...	...	...	...	...	...
case246	0,1	0,0	0,1	0,0	0,0
case247	0,0	0,1	0,1	0,0	0,6
case248	0,3	0,6	0,5	0,0	0,1
case249	0,1	0,0	0,1	0,0	0,0
case250	0,0	0,1	0,0	0,0	0,0
Qualities of representation (square cosines) of the cases by the axes					
Identifier	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
case1	0,138	0,290	0,032	0,215	0,000
case2	0,324	0,096	0,391	0,068	0,036
case3	0,001	0,027	0,115	0,213	0,153
case4	0,320	0,088	0,324	0,023	0,005
case5	0,398	0,083	0,108	0,087	0,046
...	...	...	...	...	...
case246	0,667	0,085	0,136	0,000	0,002
case247	0,094	0,183	0,187	0,025	0,452
case248	0,299	0,272	0,159	0,010	0,019
case249	0,575	0,014	0,273	0,036	0,033
case250	0,170	0,281	0,084	0,052	0,003

Figure 25- Analysis by Cases (Coordinates, Absolute Contributions, Relative Contributions)

This information will be more easily visualized through scatter plots, which will be next, and will include all 250 cases.

We then move on to analyze each principal component in more detail.

1st Principal Component

The following graphs are a visual representation of the location of the cases and the original variables across the axes 1 and 2. The table details the variables and cases with the highest coordinates, that is, the ones that are the most correlated with the 1st Principal Component.

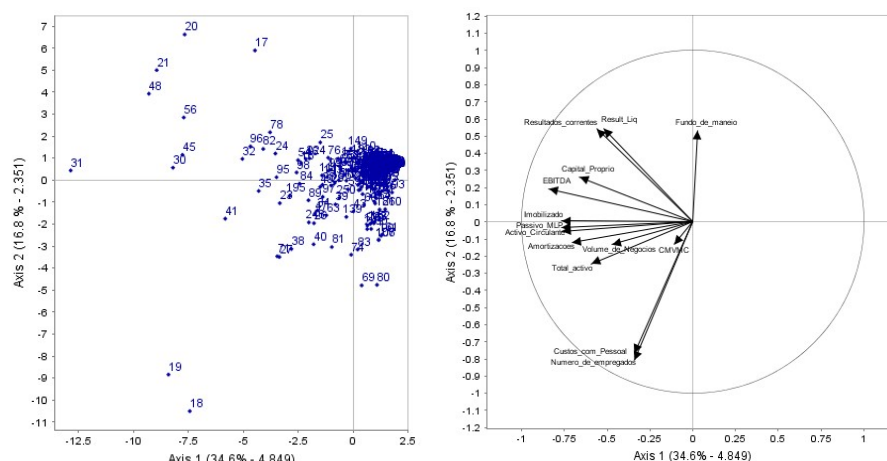


Figure 26- Cases and Original Variables across axes 1-2

Axis 1	Most Correlated	Coordinates
Variables	EBITDA	-0,843
	Passivo_MLP	-0,771
	Imobilizado	-0,769
	Activo_Circulante	-0,768
	Amortizacoes	-0,709
Cases	31 - E-REDES - DISTRIBUIÇÃO DE ELETRICIDADE, S.A.	-12,846
	48 - ANA - AEROPORTOS DE PORTUGAL, S.A.	-9,289
	21 - THE NAVIGATOR COMPANY, S.A.	-8,934
	19 - MODELO CONTINENTE - HIPERMERCADOS, S.A.	-8,398
	30 - NOS COMUNICAÇÕES, S.A.	-8,194

Figure 27- Variables and Cases most correlated with Axis 1

When we observe the variables with the highest (negative) correlation, we conclude that the 1st Principal Component mainly pertains to the EBITDA, the financial structure of the company in terms of assets and liabilities, and in particular the size of the fixed assets held by the company and its corresponding amortization costs.

Then, the 1st Principal Component opposes the companies with a large EBITDA and simultaneously a large financial structure in terms of assets and liabilities, to the companies with a relatively smaller EBITDA and financial structure.

As we can see in the first graph, most of the companies are concentrated on the positive side of the first axis. They are relatively smaller compared to a set of very few, much larger companies, namely E-Redes, ANA, The Navigator Company, Modelo Continente, and NOS. From our domain knowledge we understand these companies are particularly large due to the fixed assets associated with their activity (electrical infrastructures, airports, country-wide chains of supermarkets) and due to the size of the markets of which they are leaders.

These companies, which are on the negative side of the axis, are the ones with the largest contributions to the 1st Principal Component because they are only a very limited few, while the companies on the positive side of the axis, those relatively smaller, are much more and thus have a smaller individual contribution. This is also consistent with what we found in the exploratory data analysis, that while we are looking at the 250 largest companies, there is a small group of companies that are considerably larger than the rest.

2nd Principal Component

The following graphs are the same as in the previous section, but we will now analyze them from the perspective of the Axis 2. The table details the variables and cases with the highest coordinates, that is, those that are the most correlated with the 2nd Principal Component.

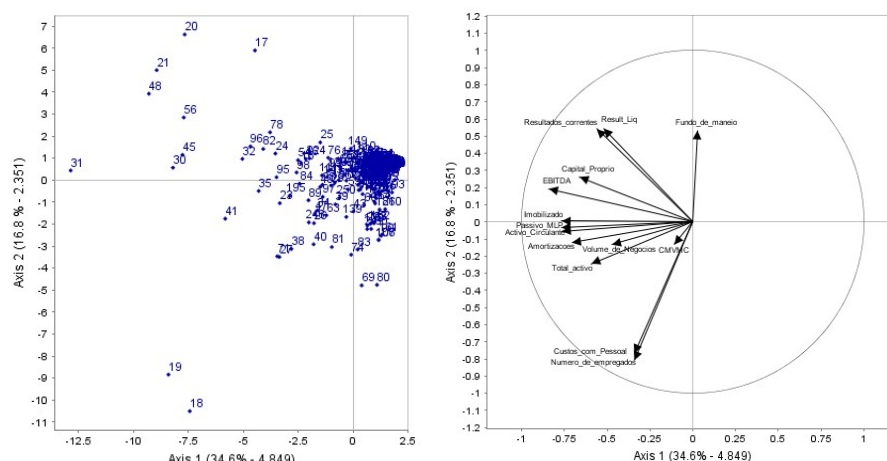


Figure 28 - Cases and Original Variables across axes 1-2

Axis 2	Most Correlated	Coordinates
Variables	Numero_de_empregados	-0,810
	Custos_com_Pessoal	-0,769
Cases	18 - PINGO DOCE - DISTRIBUIÇÃO ALIMENTAR, S.A.	-10,49966715
	19 - MODELO CONTINENTE - HIPERMERCADOS, S.A.	-8,838868544
	20 - GALP - GÁS NATURAL, S.A.	6,621
	17 - EDP - ENERGIAS DE PORTUGAL, S.A.	5,896

Figure 29 - Variables and Cases most correlated with Axis 2

The two variables that are most correlated with the 2nd Principal Component are *Numero_de_empregados* and *Custos_com_Pessoal* (negative correlation). Therefore, this component is mainly related with the number of employees of the companies and the associated costs. It is also positively correlated, to a smaller degree, with some profit variables and with working capital.

Thus, the 2nd Principal Component opposes companies that have a considerably large number of employees and associated costs, in relation to their size in terms of profit and working capital, to companies that, though large in terms of working capital and profit, have a relatively small number of employees and associated costs.

Through the graph, we can identify the 4 companies that have the largest coordinates. Pingo Doce and Modelo Continente are the most negatively correlated with this component, and these supermarket chains are also the companies with the largest number of employees of our dataset. On the positive side of the axis, we find GALP and EDP, that are the largest companies in our dataset in terms of net profit and current profit (*Result_Liq* and *Resultados_correntes*) and at the same time, have a relatively smaller number of employees and associated costs.

3rd Principal Component

Moving on to the 3rd Principal Component, the following graphs display the coordinates of the cases and the original variables across the axes 1 and 3. The table details the variables and cases with the highest coordinates, that is, those that are most correlated with the 3rd Principal Component.

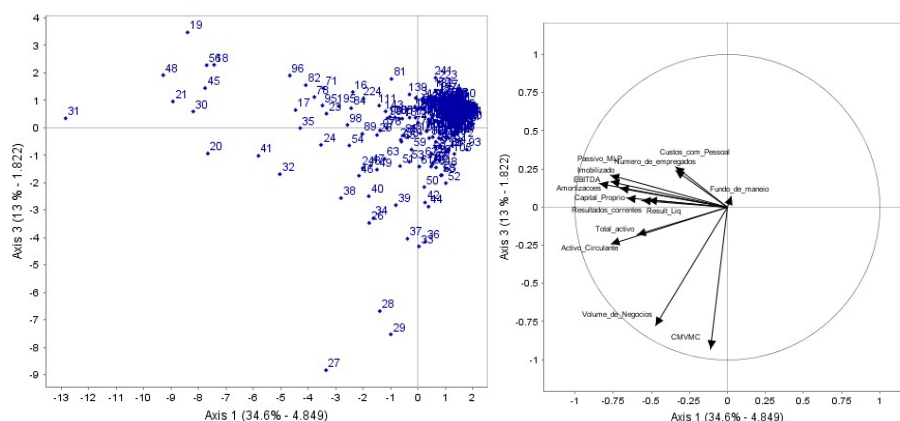


Figure 30- Cases and Original Variables across axes 1-3

Axis 3	Most Correlated	Coordinates
Variables	CMVMC	-0,932
	Volume_de_Negocios	-0,780
Cases	27 - ENDESA ENERGIA, S.A. - SUCURSAL EM PORTUGAL	-8,832
	29 - EDP GEM PORTUGAL, S.A.	-7,527
	28 - BP PORTUGAL - COMÉRCIO DE COMBUSTÍVEIS E LUBRIFICANTES, S.A.	-6,681

Figure 31- Variables and Cases most correlated with Axis 3

The variables that are most correlated (negatively) with the 3rd Principal Component are *CMVMC* and *Volume_de_Negocios*, which means that this component is mainly related to the volume of sales and the corresponding costs of the goods sold or materials consumed. All other variables are considerably less correlated.

Thus, this component opposes the companies with larger sales volume and sales costs to those with relatively smaller sales volume and costs, in relation to their overall size in terms of number of employees and associated costs, financial structure, EBITDA.

On the first graph, we can observe that most companies are concentrated around the center of the third axis, having relatively lower correlations with this component, while there are 3 companies that stand out for having a high negative correlation with this axis. They are Endesa, EDP GEM and BP. These companies are only among the 14 companies with the highest sales volume and sales costs, but they stand out because they have a particularly large sales cost and sales volume in comparison to their overall size. For instance, Petrogal (case 16) is the company with the highest sales volume and sales costs, however they have a coordinate of only 1,298 in the third Axis. This is because they have higher values in *Numero_de_empregados*, *Custos_com_Pessoal*, *EBITDA*, *Amortizacoes* and others (in comparison with the companies mentioned above), which are positively correlated with the 3rd Principal Component and thus cancel out the effect of the high Sales and Sales Costs, leading to a low coordinate.

4th Principal Component

The following graphs display the coordinates of the cases and the original variables across the axes 1 and 4. The table details the variables and cases with the highest coordinates, that is, those that are most correlated with the 4th Principal Component.

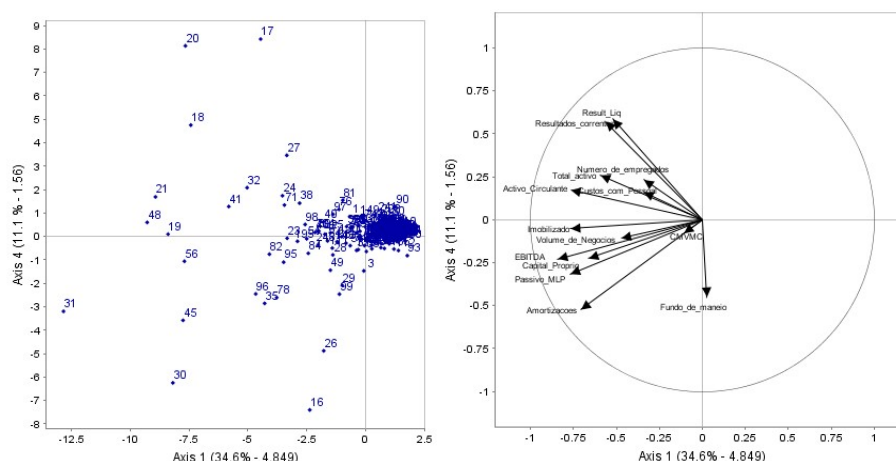


Figure 32 - Cases and Original Variables across axes 1-4

Axis 4	Most Correlated	Coordinates
Variables	Result_Liq	0,587
	Resultados_correntes	0,574
	Amortizacoes	-0,526
Cases	17 - EDP - ENERGIAS DE PORTUGAL, S.A.	8,423
	20 - GALP - GÁS NATURAL, S.A.	8,134
	16 - PETROGAL, S.A.	-7,402
	30 - NOS COMUNICAÇÕES, S.A.	-6,248

Figure 33 - Variables and Cases most correlated with Axis 4

The variables that are most correlated with the 4th Principal Component are *Result_Liq*, *Resultados_correntes* (positively correlated) and *Amortizacoes* (negatively correlated). The correlations are already considerably lower than for the previous components, but we can still conclude that this component pertains mainly to net and current profit and to the amortization costs of the companies.

Therefore, the 4th Principal Component opposes the companies that have a high net profit and current profit, but comparatively lower amortization costs, to the companies that have comparatively lower net profit and current profit, and higher amortization costs.

Looking at the first graph, we can see that this component is already not very relevant for most of the cases, that are very concentrated around the center of the axis, but there are several companies that have higher correlations, being more dispersed through either the positive or the negative side of the axis.

EDP and Galp are the two companies with the highest (positive) correlation with the fourth axis. They have high net and current profit while having relatively lower amortization costs. In fact, they are the companies with the highest net and current profits in our datasets, but they are in the 34th and 48th places in terms of amortization costs. On the other hand, Petrogal and NOS are the companies with the highest negative correlation, and they are in the 5th and 1st places in terms of amortization costs, while Petrogal has the lowest net and current profits (in fact, losses), and NOS is in the 31st place.

5th Principal Component

The following graphs show the coordinates of the cases and the original variables across the axes 1 and 5. The table details the variables and cases with the highest coordinates, that is, those that are most correlated with the 5th Principal Component.

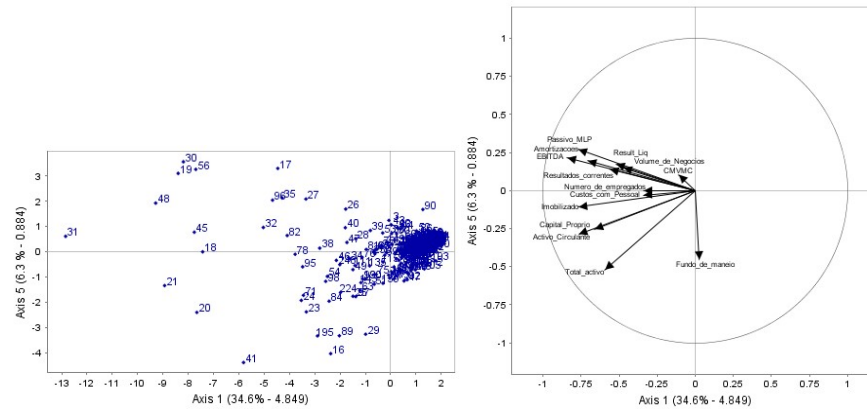


Figure 34 - Cases and Original Variables across axes 1-5

Axis 4	Most Correlated	Coordinates
Variables	Total_ativo	-0,522
	Fundo_de_maneio	-0,457
Cases	41 - MOTA-ENGIL - ENGENHARIA E CONSTRUÇÃO ÁFRICA, S.A.	-4,386
	16 - PETROGAL, S.A.	-4,035
	30 - NOS COMUNICAÇÕES, S.A.	3,563

Figure 35 - Variables and Cases most correlated with Axis 5

The variables that are most (negatively) correlated with the 5th Principal Component are *Total ativo* and *Fundo de maneo*. While the correlations are, again, considerably lower than in the first 3 components, we can say that this component mainly relates to the total assets and to the working capital of the companies.

Therefore, the 5th component opposes the companies with relatively high total assets and working capital in relation to other attributes such as liabilities, amortization costs and EBITDA, to those with relatively low total assets and working capital.

From observing the first graph, we can see that the companies are not as highly correlated with this component as in the previous ones. The most (negatively) correlated companies are Mota-Engil and Petrogal – these companies have a relatively higher total assets and working capital (particularly, Petrogal is the company with the highest working capital in our dataset), compared to their value in some other attributes that are positively correlated with this component. NOS, to a lesser extent, is positively correlated with this component, which means that this company has relatively lower total assets and working capital in relation to some other attributes. Since this is already the 5th Principal Component, that explains about 6% of the variance, these oppositions are already not as evident or relevant as in the first components.

ii. Conclusion

Overall, this analysis was able to identify the main components of this dataset, that explain a large part of the differences between the 250 largest Portuguese companies through several economic, financial and activity variables. Through the first 3 components, the companies were differentiated according to their EBITDA and financial structure, to their number of employees and associated costs, and to their sales volume and associated costs. The 4th and 5th components, while still retained according to several criteria, produced less meaningful oppositions, with lower correlations to the variables and cases, but they pertained to Net and Current profits in opposition to Amortization Costs, and to Total Assets and Working Capital, respectively. These are, therefore, the main dimensions through which we can compare and distinguish the largest Portuguese companies.

iii. Principal Component Analysis of the Ratio Variables

Principal Component Analysis was also performed on the ratio variables, with the purpose of applying a clustering model to the obtained Principal Components. While the Cluster Analysis will be presented in the following section, we now present a summary of the PCA results.

Considering the variables from the economic and financial ratios as active variables and all the others as supplementary variables, we performed normed Principal Component Analysis, given that our variables have different scales.

We initially included all cases, but the case 43 - AMERICAN AIRLINES, INC - SUCURSAL EM PORTUGAL stood out as being very distant from the other cases in every axis. The results of this PCA can be found in the appendix. Considering the status of this case as a severe outlier, shown both in the EDA as well as in this PCA, we decided that for the purpose of the Clustering, it would be preferable to perform a PCA excluding this case, as it could lead to a bias in the Clusters. The following results were obtained.

Variances of axes (eigenvalues)

Cloud variance (matrix trace) : 10

Axis	Variance of the axis (eigenvalue)	% of explained variance	Cumulated % of explained variance	Retained according to Kaiser's criterion
1	3,554	35,5	35,5	X
2	1,991	19,9	55,5	X
3	1,039	10,4	65,8	X
4	1,003	10,0	75,9	X
5	0,907	9,1	84,9	
6	0,702	7,0	92,0	
7	0,478	4,8	96,8	
8	0,257	2,6	99,3	
9	0,068	0,7	100,0	
10	0,000	0,0	100,0	
Total	10,000	100,0	100,0	

Variance of the axes - decreasing curve

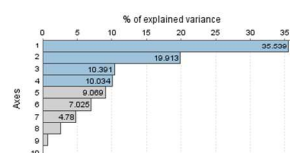


Figure 36 - Variance of the axes

By Kayser's criterion, 4 axes are to be considered as Principal Components, and by Pearsons' criterion we could consider 4 or 5 Principal Components (75,9% and 84,9% of explained variance, respectively), whereas Cattle's criterion points out to 3 Principal Components. Since 4 Principal Components seems to be the consensual figure, we will consider it.

The following figure depicts the coordinates of the cases and the ratio variables across the first 2 axes. Since this is a normed PCA, the coordinates also correspond to the correlations. The graphs for the remaining selected axes can be found in the appendix X.

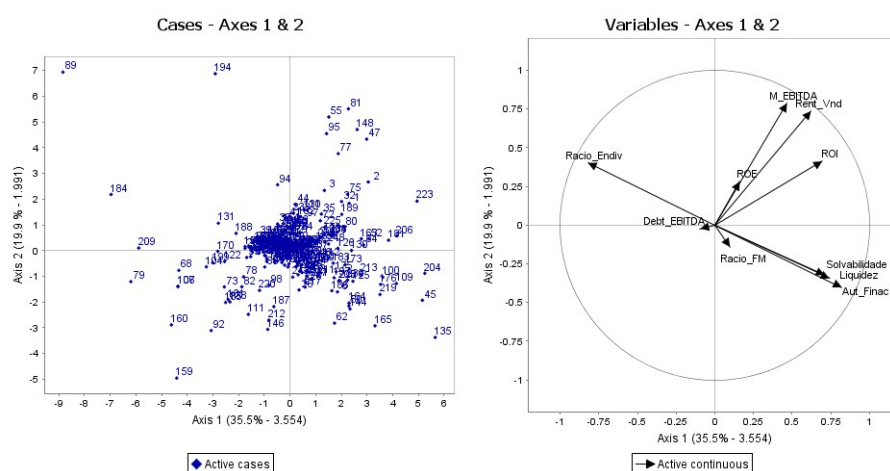


Figure 37 - Cases and Ratio Variables across axes 1-2

From this graph, we can observe that the 1st Principal Component is most highly correlated with Racio_Endiv (negatively) and with Aut_Finac, Solvabilidade, Liquidez and ROI (positively). Then, this Principal Component mainly opposes companies that have a challenging financial situation and low profitability to companies with a high profitability and strong financial situation.

The 2nd Principal Component is most highly correlated with Rend_Vnd and M_EBITDA (positively). Therefore, this Principal Component mainly opposes

companies that have a high economic profitability, to those that have a low economic profitability.

When analysing the coordinates of the cases (figure 37), the case 89 - TEIXEIRA DUARTE - ENGENHARIA E CONSTRUÇÕES, S.A. stands out as being very distant from all other cases, with much higher coordinates, in both axes. This means that this company has a poor financial situation, while still having a comparatively good profitability. In fact, this company has a high debt ratio and low solvency, and while the profitability ratios are negative, they are very close to zero. The company 194- RENOVÁ - FÁBRICA DE PAPEL DO ALMONDA, S.A. also stands out as being in a similar, though more moderate, situation.

We now move on to the cluster analysis of the ratio variables, and then to the cluster analysis of the principal components we just presented.

2.2 Cluster Analysis

i. Cluster Definition

Clustering is about finding groups in our dataset, organizing cases in homogenous groups. In cluster analysis we want to find groups of cases that are similar and put those in evidence, putting together what is similar and separating what is different.

In our data set we have the 250 largest Portuguese companies by turnover, described by financial indicators. With cluster analysis we are interested in finding groups of companies with the same typology or profile: organize companies in classes such that companies of the same class are homogeneous and that the classes are relatively distinct from each other.

There are many clustering models, the most popular ones being Partition Models and Hierarchical Models: in the Hierarchical Models classes are organized in a nested structure (each element belongs at least to one class), whereas in Partition Models classes are organized in disjoint structures which together cover the whole data set (each element belongs just to one class).

To organize our data set into classes, we used the Hierarchical Model. This model has several advantages over the Partition Model, of which we would like to highlight the fact that it does not require a previous definition of a number of classes and it provides a visual representation of the classes in a hierarchy, making it easier to visually identify clusters and how they relate with each other.

The identification of clusters depends on the data set, the type of variables, the choice of variables, the comparison (dissimilarity) measure between cases and the clustering method (the comparison/dissimilarity measure between classes or aggregation index), so before the cluster procedure begins there are choices to be made.

Our data set is composed of 250 cases, and 19 attribute variables. Even though the companies in the data set are the largest companies in Portugal, as described in the EDA, they present different sizes, with the largest company with a turnover around € 14 thousand million and the smallest around € 160 million. The variables themselves present very different scales, with some variables in the order of thousand million and others in the order of thousands. Consequently, our first choice was to standardize values, so not to give more importance to some variables over others.

Regarding the choice of variables, considering that we want to compare companies from an economic and a financial point of view, ratios rather than absolute values provide a better way to compare companies from different sectors or sizes. Having that in mind, in order to form relevant clusters of companies, we computed the following economic and financial variables⁵: *Racio_FM* (Working Fund Ratio), *Liquidez* (Liquidity ratio), *Aut_Finac* (Financial Autonomy), *Racio_Endiv* (debt-to-total-assets ratio), *Solvabilidade* (Solvency Ratio), *Debt_EBITDA* (Debt to EBITDA

⁵ Full description of the variables in section 1.1 – Data.

Ratio), *M_EBITDA* (EBITDA Margin), *Rent_Vnd* (Profit Margin), *ROI* (Return on Investment) and *ROE* (Return on Equity). We considered these variables together with *PMR_dias*, *Stock_Turnover_per* and *PMP_dias*, so in total 13 variables were considered.

Our third choice concerned the comparison measure between cases, and our preference fell upon the dissimilarity measure Squared Euclidian Distance. Lastly, we defined the Ward's Linkage as the aggregation index of choice: we tried different aggregation indices, and the Ward's Linkage was the one with a clearer definition of clusters.

Having defined our preferences, we used the SPSS software to identify the clusters and to plot the Hierarchical Clustering. One of the companies, *American Airlines, Inc - Sucursal Em Portugal* appeared as an outlier⁶, (having a small balance sheet and a relatively high turnover, benefiting from the structure of its parent company), affecting the plotting of the cluster, taking us to decide to leave this company out of the clustering analysis, as it has a class of its own.

Bearing in mind that the data set at hand has 250 companies (249, considering the exclusion of *American Airlines, Inc.*), plotting such a number of observations would always be a challenge, and the results obtained were the following (SPSS file of the clustering attached):

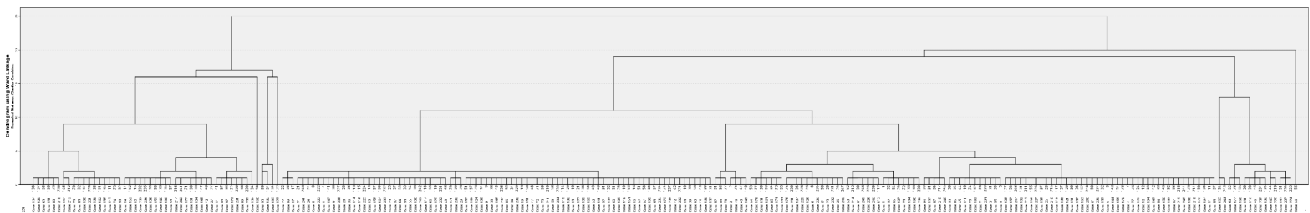
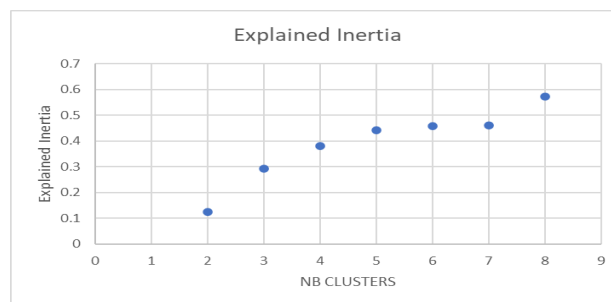


Figure 38 - Dendrogram

From the representation of the Clusters, we can identify between 3 or 4 clusters of companies: as we can see from the plot, there are 3 well defined groups of companies, and we can also consider another one, formed with a single company (*Fratelli Cosulich, Unipessoal, S.A. (Zona Franca Da Madeira)*).

To help us determine the number of clusters to consider, we computed additional indices, namely Explained Inertia index, the Calinsky-Harabaz index and the Silhouette index. These indices were considered for the range from 2 to 8 clusters, whose results are plotted in the following figures:



⁶ In Exploratory Data Analysis, this company was identified as a severe outlier in 4 ratio variables. It also stood out as an outlier in every axis of the Principal Component Analysis.

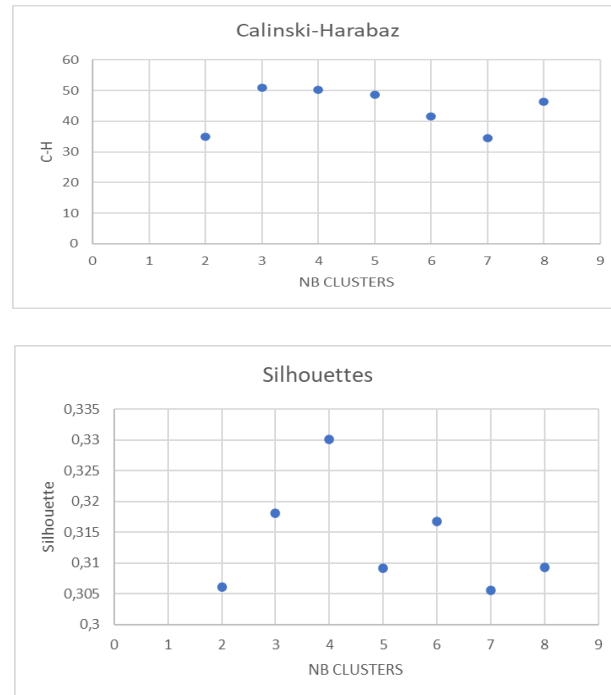


Figure 39 - Scatter Plots: Explained Inertia, Calinsky-Harabaz and Silhouettes

From the explained inertia plot, we can see that the explained inertia increases up to 5 clusters, and then stabilizes, increasing again from 7 to 8 clusters. The Explained Inertia index is a ratio of the Between Inertia and Total Inertia, indicating how well the classes separated. From the plot we can see that there are considerable gains in separation from 2 to 3 clusters, from 3 to 4 still increases but not as much. From 4 to 5 clusters the gain is relatively small, stabilizing after that. So, from this index, we can take that 3 or 4 clusters may be considered.

The Calinsky-Harabaz index is a weighted ratio of the Between dispersion (which we want to be high) and the Within dispersion (which we want to be low), so we are looking for a peak in this index. From the plot, we can see that the index peaks at 3 clusters, maintaining a similar value at 4 clusters and then it starts decreasing. So, 3 or 4 cluster seem to be an option considering this index.

Finally, the Silhouette index measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation), ranging from -1 to +1: we are looking for high values, that are indicative that the objects are well matched to their own cluster and poorly matched to the neighbouring classes. Looking at the Silhouette Plot it's clear that the silhouette index increases up to 4 clusters and then it decreases, indicating that 4 clusters is a division of clusters to consider.

Given all the above, we conclude that the division in 4 clusters seems to be appropriate, consisting in 3 clusters of companies and one cluster with just one company. The red dashed line in the figure below indicates the cutting level and the corresponding 4 clusters.

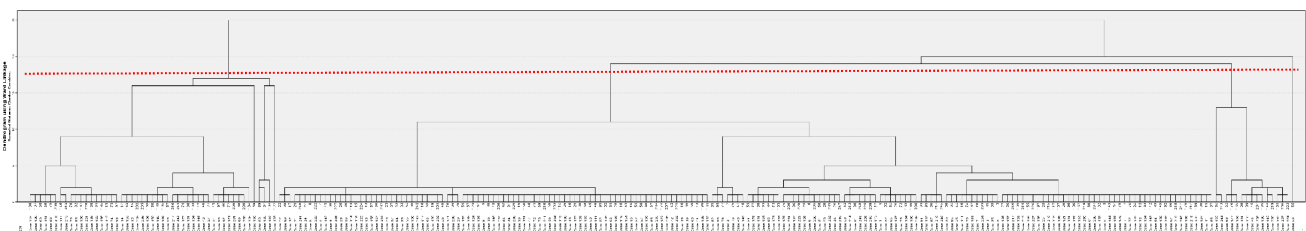


Figure 40 - Dendrogram in 4 clusters

2.2.2 – Partition Interpretation

ii. Partition Interpretation

Now that we have identified a partition of 4 clusters to our data set, let's analyse each of those groups with the help of the Case Summary table for the 4 clusters partition.

Case Summaries										
Mean										
Ward Method	Racio_FM	Liquidez	Aut_Finac	Racio_Endiv	Solvabilidad	Debt_EBITDA	M_EBITDA	Rent_Vnd	ROI	ROE
1	4,28105	1,29902	,34355	,65645	,70717	18,25318	,10650	,06336	,09636	,25056
2	1,76904	,95507	-,04283	1,04283	,18712	66,97194	-,00299	-,04680	-,08294	-,58771
3	42,96621	3,35486	,74716	,25284	3,91260	3,12490	,15646	,12923	,13524	,14878
4	,87290	,44450	,10603	,89397	,11860	4,00476	,74426	,55018	,16502	,93674
Total	6,10346	1,35175	,29087	,70913	,79557	26,87184	,09052	,04760	,06370	,08223

Figure 41 - Case Summaries

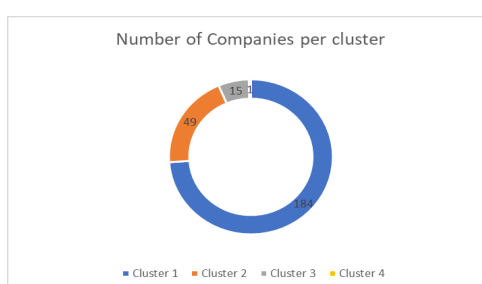


Figure 42 - Number of companies in each cluster

Considering the variables selected for clustering, the clusters should select groups of companies according to their economic and financial health.

The first cluster that calls our attention is Cluster 3. Companies in this Cluster (15 companies) have a strong financial health and a strong economic performance. Companies in this cluster are very well capitalized (average Financial Autonomy above 74%), high liquidity (mean assets to be converted in short term are 3 times over the mean short-term responsibilities), low debt and a strong capability to repay debt with generated EBITDA. From an economic point of view these are companies with strong margins and high Return on Investment. Given that these are companies with strong levels of equity, despite its good economic performance, Return on Equity is rather moderate.

On the opposite side, with 49 companies, is Cluster 2: companies in this cluster have a poor financial health and a weak economic performance. Companies in this cluster have, on average, negative equity (technical bankruptcy), high levels of debt, and low liquidity. On the economic side, its performance is frail, with negative margins and returns.

Cluster 1 gathers companies with an intermediate financial and economic performance. This cluster, with 184 companies (the largest cluster), has companies with moderate levels of capitalization (mean Financial Autonomy around 34%) and adequate levels of liquidity (mean assets to converted in short term are slightly over the mean short-term responsibilities). From an economic point of view, these are companies with relatively good margins (although the not as high as in cluster 3), and that have the highest Return on Equity (mean ROE over 25%, due to relatively good business margins and moderate capitalization).

Lastly, we have Cluster 4 with just one company. This company does not fit any of the other clusters, as it has relatively low levels of equity and liquidity but an outstanding economic performance (quite large business margins), and a ROE near

100%. This is in fact an outlier (we already excluded one company as an outlier for the purpose of clustering, nevertheless another one is clear).

To illustrate these 4 clusters, figure 43 below presents boxplots for the Financial Autonomy, evidencing the different types of companies in each class (boxplots for other variables appended). As we can see from the figure below, Cluster 2 has low capitalization companies, being clear that there are several outliers, with very negative capitalization levels. Cluster 3, on the other hand, has very strong levels of Financial Autonomy (close to 100%), whereas Cluster 1 has an intermediate level of capitalization. Lastly, Cluster 4 has just the one company, with a low level of Financial Autonomy.

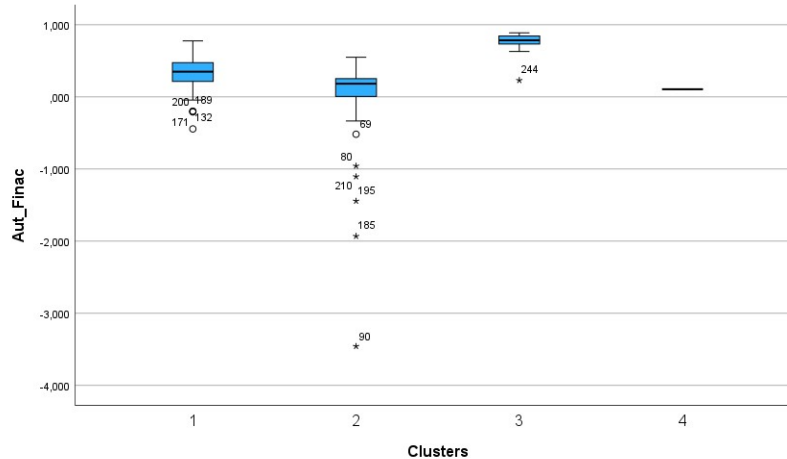


Figure 43 – Boxplots: Financial Autonomy

From the economic point of view, we have the same scenario, where Cluster 3 has large operating margins, Cluster 2 with roughly 50% of the companies operating with negative margins, and, Cluster 1 in an intermediate position, evidencing also several outliers with strong margins. Cluster 4, this time around, presents unusual levels of EBITDA margins.

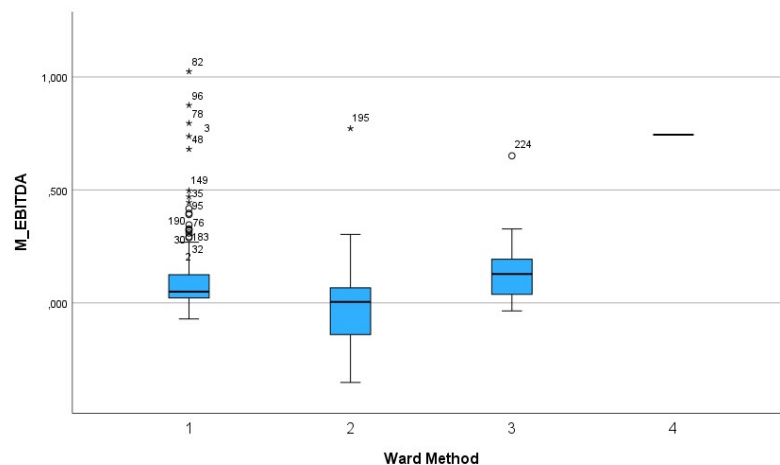


Figure 44 - Boxplots: EBITDA Margin

iv. Non-Hierarchical Clustering

Besides Hierarchical Clustering, we also tried to cluster our data set with a different model, namely a Partition Model, to check if a better division of classes could be achieved. To that effect we conducted a Cluster Analysis using K-Means algorithm. Selecting the same variables as in Hierarchical Clusters and defining 4

clusters, we ended up with almost the entire data set in one cluster, and the remaining 3 clusters with one and five companies. To reach a partition where we have at least 3 well defined groups of companies, we had to select 7 clusters, and even so more than 70% of the companies were in the same group (output from SPSS attached).

Given the exposed, we consider that the Non-Hierarchical model wasn't the best fit for our data set, when comparing with the results obtained by the Hierarchical Model.

Nevertheless, analysing the ANOVA table for the clustering with 7 classes we concluded that the variables that best separate the groups are Debt_EBITDA, ROE and Racio_FM.

v. Hierarchical Clustering with Principal Component Analysis

Lastly, we are going to analyse the clustering of our data set applying the hierarchical model not to the original data but to its principal components, that is, we will chain Clustering Analysis to the Principal Component Analysis of the ratio variables, using the Principal Components that have already been presented in the PCA section of this report.

Considering Ward's Linkage as the aggregation index, the following Dendrogram was obtained.

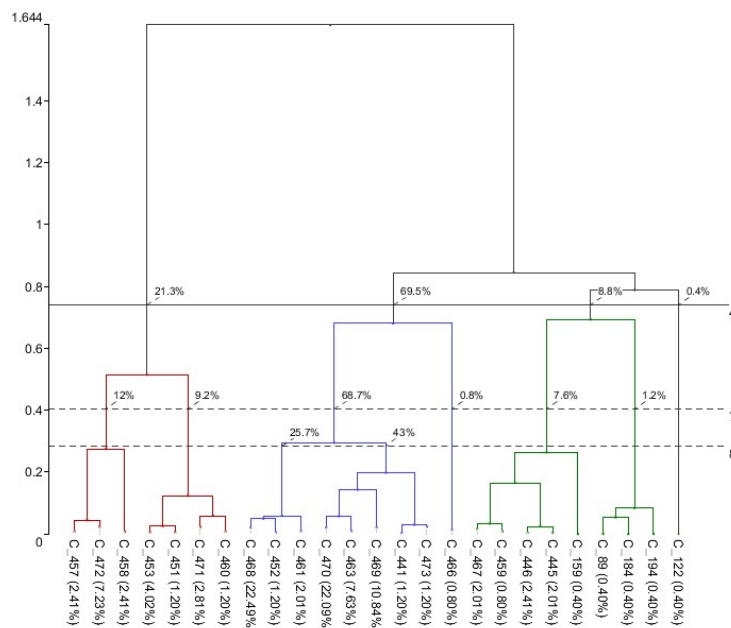


Figure 45 - Dendrogram: PCA

The obtained results suggest the partition of the data set in 4, 7 or 8 clusters. Naturally the partition in 4 clusters catches our eye, as it is the same partition obtained in clustering from the original variables. In this partition we also have a cluster with most of the observations, two smaller ones and one cluster with just one observation.

Moreover, looking at the several indices to the determination of clusters, we could consider between 6 and 8 clusters:

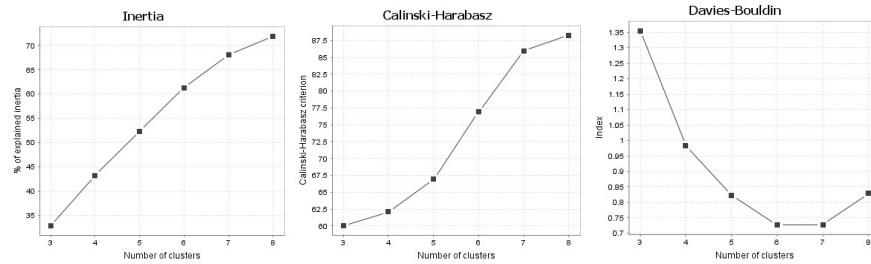


Figure 46 - Indices for Determination of Number of Clusters

From the analysis of the above indices, the 7-cluster partition is suggested: the percentage of total inertia grows at relatively large rates until it reaches the 7th cluster; the Calinski-Harabasz index peaks at 8 clusters, but the difference to 7 clusters is relatively small; Davies-Bouldin index is low at 6 and 7 clusters. So, 7 cluster seems to be most consensual partition.

However, looking at the results of such partition (files from SPAD software appended), we verified that further partition was made not on the main class (where eventual further partition could be informative, as it would be breaking down a relatively large class), but on the smaller classes. As we will see later in this report, the smaller classes are very well identified (they are somewhat extreme: strong and frail companies), consequently the benefits of the analysis of further partition are rather small.

Given the above, we can interpret the suggestion to 7 cluster from the above indices as symptomatic that in both groups of frail and strong companies there are different levels of frail and strong, respectively, that could be clustered. However, this division of classes is not all that interesting for our analysis, since these are already relatively small classes and that further division wouldn't be all that informative, considering that these are already extreme classes (for the purpose of our analysis, there's no substantial gain in differentiate extremes). A totally different issue would be if further partition could be done in the main class, where there are almost 70% of the companies, as it would be interesting to make further distinctions in the group of companies that are not in extreme classes. Nevertheless, this finding is in itself informative, as it indicates that the companies in the "middle of the range" class are relatively similar (the Within Variance should be low in this class, which is what we want to achieve when clustering), so that no further clustering is suggested. Further ahead on our analysis we will present graphical representation that illustrates these findings.

Having said that, and considering that the 4-clusters partition allows a comparison with the results obtained in clustering the original data and that it seems to be a natural partition, we will consider it.

Partition in 4 clusters

Informations on the clusters

Cluster	Before consolidation			After consolidation		
	Count	Percentage	Inertia	Count	Percentage	Inertia
1	53	21,285	1,184	57	22,892	1,573
2	173	69,478	1,765	171	68,675	1,545
3	22	8,835	1,362	20	8,032	0,967
4	1	0,402	0,000	1	0,402	0,000
Overall	249	100,000	4,311	249	100,000	4,085

Figure 47 - Information on Clusters

In a 4-Cluster partition (after consolidation), we will have one main cluster (cluster 2) with 171 companies (68,7% of the companies), followed by two smaller ones (57 companies in cluster 1 and 20 in cluster 3) and a final cluster with just one company. So, in this partition we have a similar structure as the one obtained with clustering on the original data.

Looking at the quality indicators, on figure 48 below, we can see that in this partition the *Within Cluster Variance* is higher than the *Between Cluster Variance*,

which is symptomatic that further partition could be obtained, as our classes may not be sufficiently homogeneous and distinct from each other. However, as explained above and as figure 53 below shows, the within cluster variance comes from clusters 3 and 1, which are extremes and relatively small in size, so the benefits of further partition would be small.

Quality indicators			
Name	Before consolidation	After consolidation	
Within cluster variance	4,311	4,085	
Between cluster variance	3,277	3,503	
Between variance rate (η^2)	43,187	46,161	
Calinski-Harabasz (pseudo F) criterion	62,079	70,019	
Davies-Bouldin's index	0,982	0,884	

Consolidation of the partition			
Iterations	Total variance	Between cluster variance	Ratio
0	7,588	3,277	0,432
1	7,588	3,499	0,461
2	7,588	3,503	0,462

Figure 48 - Quality indicators and variance

Notwithstanding, the partition in 4 clusters explains almost half (46%) of the differences between the groups.

Even though our cluster analysis was based on Principal Components, it is possible to obtain a direct characterization of the clusters in terms of the original variables. This is obtained by chaining “Characterization of the Clusters of the Typology” to our Cluster Analysis. The results obtained were the flowing:

Classe : Cluster 1/4 (Count : 57 - Percentage : 22.89%)			Classe : Cluster 2/4 (Count : 171 - Percentage : 68.67%)		
Characteristic variable	Mean in the partition	Global mean	Characteristic variable	Mean in the partition	Global mean
Solvabilidad	2,060	0,796	Rent_Vnd	0,034	0,048
Liquidez	2,218	1,352	M_EBITDA	0,069	0,091
ROI	0,205	0,064	Liquidez	1,154	1,352
Rent_Vnd	0,164	0,048	Solvabilidad	0,485	0,796
M_EBITDA	0,233	0,091			
Aut_Finac	0,592	0,291			
Racio_Endiv	0,408	0,709			
Classe : Cluster 3/4 (Count : 20 - Percentage : 8.03%)			Classe : Cluster 4/4 (Count : 1 - Percentage : 0.40%)		
Characteristic variable	Mean in the partition	Global mean	Characteristic variable	Mean in the partition	Global mean
Racio_Endiv	1,421	0,709	Debt_EBITDA	3397,263	26,872
Solvabilidad	-0,111	0,796			
ROE	-1,514	0,082			
Liquidez	0,566	1,352			
M_EBITDA	-0,123	0,091			
Rent_Vnd	-0,164	0,048			
Aut_Finac	-0,421	0,291			
ROI	-0,231	0,064			

Figure 49 - Characteristic Active Variables

Comparing the Characteristic Variables of each cluster with the respective global means, we can characterize each cluster in terms of the original variables.

In Cluster 1, the economic and financial ratios have values considerably higher than the mean values, meaning that this cluster gathers companies with a robust economic and financial structure. To paint a picture of the type of companies that are in this cluster, we can use the “centroid company” of the cluster: it is a company with a very high financial autonomy (around 60%), high liquidity, high EBITDA Margins (EITDA is 23,3% of the turnover), high Return on Investment (20,5%) and low leverage (low debt-to-total-assets ratio). All round, these are very strong values, symptomatic of very healthy companies.

In Cluster 3 we have very frail companies, from an economic and financial point of view. These are companies with high levels of debt and low levels of Equity (the mean company is in technical bankruptcy), liquidity and margins (among others).

Cluster 2 is the biggest cluster, characterized by having economical and financial ratios slightly lower than the mean global values.

Cluster 4 has just one company, which is an outlier (*Fratelli Cosulich, Unipessoal, S.A. (Zona Franca Da Madeira)*), characterized by having an oddly large value of Debt to EBITDA.

The following figure gives a graphical representation of the cluster in terms of the Characteristic Variables, where the blue columns represent variables with values higher than the mean global values and the red columns represent variables with values lower than the mean global values.



Figure 50 – Bar plots for Characteristic Active Variables

These clusters can also be characterized by the supplementary variables, as show in the following tables.

Classe : Cluster 1/4 (Count : 57 - Percentage : 22.89%)			Classe : Cluster 3/4 (Count : 20 - Percentage : 8.03%)		
Characteristic variable	Mean in the partition	Global mean	Characteristic variable	Mean in the partition	Global mean
EBITDA	95 151 939	44 844 815	Custos_com_Pessoal	133 163 561	43 909 760
Resultados_corrientes	59 657 997	24 813 337	PMP_dias	81	47
Result_Liq	43 765 919	18 458 092	Numero_de_empregados	3 678	1 453
CMVMC	180 831 130	454 955 959	Resultados_corrientes	-51 363 310	24 813 337
Custos_com_Pessoal	25 235 484	43 909 760	Result_Liq	-49 260 067	18 458 092
PMP_dias	32	47	EBITDA	-37 693 909	44 844 815

Classe : Cluster 2/4 (Count : 171 - Percentage : 68.67%)		
Characteristic variable	Mean in the partition	Global mean
CMVMC	582 059 022	454 955 959
Volume_de_Negocios	729 704 367	607 636 155

Figure 51 - Characteristic Supplementary Variables

Looking at the results, we notice that, whereas the Characteristic Active Variables that illustrate the clusters are an ensemble of economic and financial ratios, the corresponding Supplementary Variables are essentially economic.

Unsurprisingly, companies in Cluster 1 have considerably larger mean values in EBITDA, Net Income and Income from continuing operations before interests and tax, and considerably smaller mean values in Cost of goods sold and materials consumed, Employee benefits expenses and Average payment period (days). These are all indicators of companies with strong economic health.

Cluster 3 considers almost the same variables but in the opposite direction: considerably higher values in Cost variables and lower values in Income variables.

Cluster 2 is represented by Turnover and Cost of goods sold and materials consumed, both variables slightly higher than the global mean.

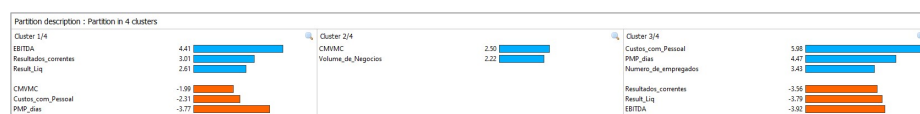


Figure 52 -Bar plots for Characteristic Supplementary Variables

Lastly, to end our Hierarchical Clustering with Principal Component Analysis, we will inspect the projection of the clusters in the Principal Components plain.

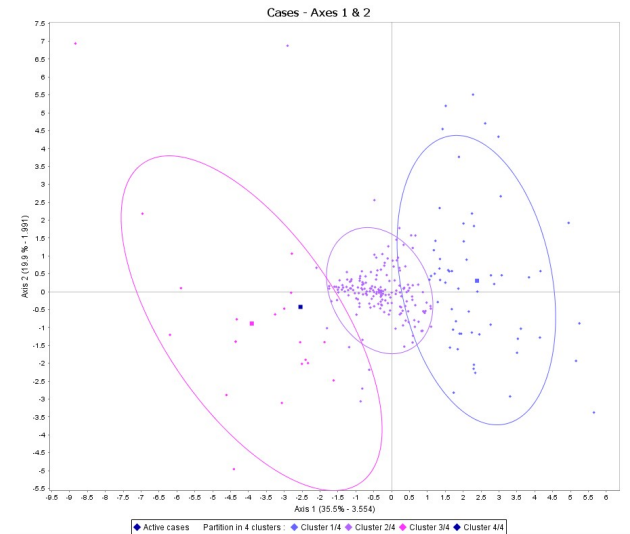


Figure 53 - Projection of the clusters: 4 clusters

As we can see from figure 53, Axis 1 (the first Principal Component) opposes well Cluster 3 (frail companies, on the negative side of axis 1) and Cluster 1 (strong companies, on the positive side of axis 1).

It also should be noticed that even though Cluster 2 comprises almost 70% of the companies, is a very compact cluster: all observations are very close together right in the middle of axis 1 and axis 2, confirming the statute of “middle of the range”. On the other hand, classes 1 and 3 have larger dispersion, despite having a considerably smaller number of entities. In both frail and strong companies’ clusters we have very different levels of frail and strong, respectively, nevertheless, that is not a so interesting distinction to our analysis. For instance, the distinction between a frail company and a much frailer one has little value added: they are both poor quality companies; that’s what defines them, rather than the degree of frailness. Still, even with 4 clusters, it’s possible to identify companies in these 2 clusters, that are far away from the centre, like companies 89 and 194, for cluster 3 and companies 81, 55 or 135 in cluster 1.

Having this in mind we can understand why the cluster analysis suggested 7 clusters and that, to our purpose, 4 clusters presents a better option to analyse the dataset, as the figure below, representing the scatter plot for 7 clusters, illustrates.

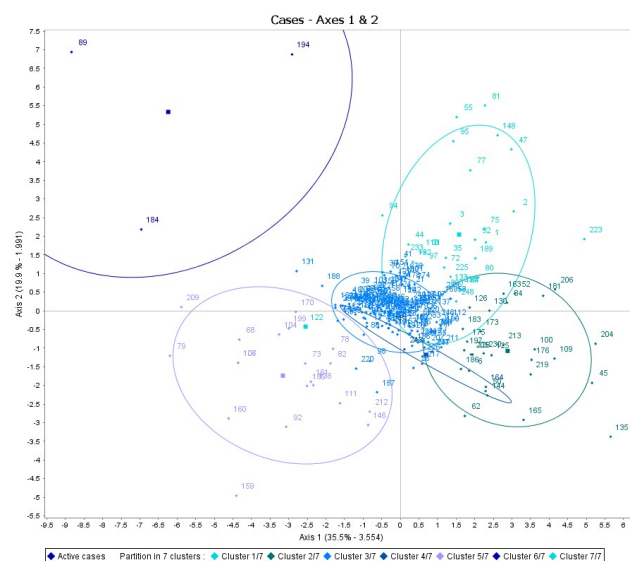


Figure 54 - Projection of the clusters: 7 clusters

vi. Final considerations

In short, the data set of the 250 largest companies by turnover in 2022 has 3 main clusters of companies:

- two smaller clusters in number of entities but large in dispersion, consisting, on one hand, of very strong economic and financial companies and, on the other hand, frail companies (extreme clusters).
- a large cluster of companies in number of entities but small in dispersion, consisting of companies with a relatively stable economic and financial situation.

We have also identified 2 outliers, one of them left out of clustering with the purpose of obtaining a better representation of the clusters and the other one identified in one cluster.

This way, we can state that even though the *250 largest companies by turnover* data set has companies from very different sectors, locations and dimensions, the large majority of companies have a similar financial economic and financial structure, namely a relatively stable economic and financial situation and exhibiting relatively high Return on Equity. There is also a small number of companies with exceptional economic and financial performance, so, all in all, around 79% of the companies in the dataset are healthy companies, from an economic and financial point of view.

However, as big doesn't necessarily mean good, our cluster analysis identified around 50 companies with a poor economic and financial performance.

The aim of our analysis was to identify the main groups of companies, which lead us to 3 main clusters. This partition in 3 clusters still has considerable Within Cluster variance, mainly from the 2 extreme clusters, which we did not explore as it wouldn't benefit the purpose of our analysis. However, further partitions should be considered if there is an interest in studying in more detail these extreme clusters.

3 Conclusion

Our report presented a comprehensive univariate, bivariate and multivariate analysis of the Top 250 Portuguese Companies by turnover in 2022.

The univariate analysis we performed allowed us to gain a general insight into the dataset. We found that, overall, it's a well composed and clean dataset, with a moderate number of missing values and/or zeros.

Most of the variables are numerical continuous and present a positively skewed distribution.

It became clear in our analysis that there is, among the top 250, a small group of companies that are the largest of the largest: around 5% of the dataset are companies with a turnover between € 2 billion and € 15 billion, while the remaining 95% of the dataset have a turnover between € 160 million and € 2 billion. This group of the largest of the largest generally translates into outliers across the variables.

The top 250 Portuguese companies are geographically concentrated: about 70% of are located in Lisboa or Porto.

In the bivariate analysis, we observed that some of the numerical variables have a significant level of correlation between them, particularly among the economic variables, as was expected.

We also confirmed that, as expected, the type of economic activity (represented by CAE_agr) has a significant effect on the economic and financial structure of the companies, as seen with the Kruskal-Wallis's test.

We also analysed 10 new numerical continuous ratio variables, finding out that there is one company that's a severe outlier in several variables, and that there are some strong correlations between a few pairs of variables.

In the multivariate analysis, we performed Principal Component Analysis and Cluster Analysis on the dataset.

The Principal Component Analysis identified 5 Principal Components through which we can compare and distinguish the largest Portuguese companies: the 1st component was most correlated with EBITDA and financial structure variables, which can then be considered the most important attributes for comparing and distinguishing the companies. The 2nd and 3rd differentiated the companies through number of employees and associated costs, and sales volume and associated costs, respectively. The 4th and 5th components, while still retained through several criteria, were less relevant than the previous, and pertained to net and current profits and amortization costs, and to total assets and working capital, respectively.

The Cluster Analysis, which used the ratio variables, identified 3 main clusters: two small and disperse clusters consisting of very economically and financially strong companies and of economically and financially frail companies, respectively, and one large and compact cluster of companies with a relatively stable economic and financial situation. There were also 2 outlier companies.

From this analysis, it became clear that, while the largest 250 companies have various sectors, locations, and dimensions, a large majority of them are in a similar and relatively positive economic and financial situation. There is also a small number of companies with exceptionally positive economic and financial performance, and on the other hand, around 20% of the companies that were identified as having a poor economic and financial performance.

Appendix

I - Kolmogorov-Smirnov test for Numerical Variables

KS Test for CAE_agr against normal distribution:
Statistic: 0.9960
P-value: 0.0000
The CAE_agr variable does not follow a normal distribution.

KS Test for Imobilizado against normal distribution:
Statistic: 0.9960
P-value: 0.0000
The Imobilizado variable does not follow a normal distribution.

KS Test for Activo_Circulante against normal distribution:
Statistic: 1.0000
P-value: 0.0000
The Activo_Circulante variable does not follow a normal distribution.

KS Test for Total_activo against normal distribution:
Statistic: 1.0000
P-value: 0.0000
The Total_activo variable does not follow a normal distribution.

KS Test for Capital_Proprio against normal distribution:
Statistic: 0.9200
P-value: 0.0000
The Capital_Proprio variable does not follow a normal distribution.

KS Test for Passivo_MLP against normal distribution:
Statistic: 0.9200
P-value: 0.0000
The Passivo_MLP variable does not follow a normal distribution.

KS Test for Fundo_de_manejo against normal distribution:
Statistic: 0.7320
P-value: 0.0000
The Fundo_de_manejo variable does not follow a normal distribution.

KS Test for Numero_de_employees against normal distribution:
Statistic: 0.9640
P-value: 0.0000
The Numero_de_employees variable does not follow a normal distribution.

KS Test for Volume_de_Negocios against normal distribution:
Statistic: 1.0000
P-value: 0.0000
The Volume_de_Negocios variable does not follow a normal distribution.

KS Test for CMVMC against normal distribution:
Statistic: 0.9320
P-value: 0.0000
The CMVMC variable does not follow a normal distribution.

KS Test for Custos_com_Pessoal against normal distribution:
Statistic: 0.9800
P-value: 0.0000
The Custos_com_Pessoal variable does not follow a normal distribution.

KS Test for Amortizacoes against normal distribution:
Statistic: 0.9760
P-value: 0.0000
The Amortizacoes variable does not follow a normal distribution.

KS Test for EBITDA against normal distribution:
Statistic: 0.8920
P-value: 0.0000
The EBITDA variable does not follow a normal distribution.

KS Test for Resultados_correntes against normal distribution:
Statistic: 0.8280
P-value: 0.0000
The Resultados_correntes variable does not follow a normal distribution.

KS Test for Result_Liq against normal distribution:
Statistic: 0.8160
P-value: 0.0000
The Result_Liq variable does not follow a normal distribution.

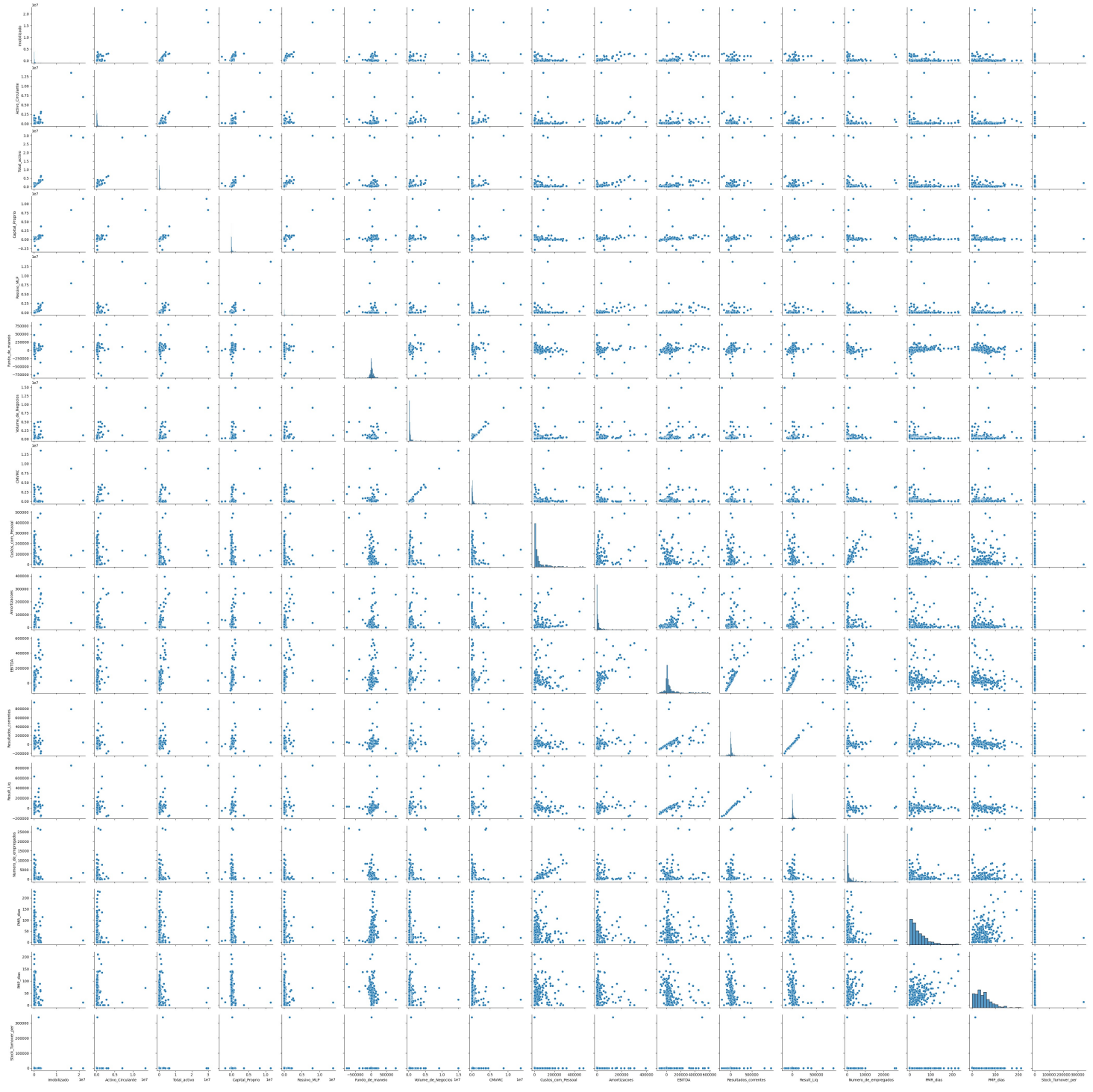
KS Test for PMR_dias against normal distribution:
Statistic: 0.8890
P-value: 0.0000
The PMR_dias variable does not follow a normal distribution.

KS Test for Stock_Turnover_per against normal distribution:

Statistic: nan
P-value: nan
The Stock_Turnover_per variable follows a normal distribution.

KS Test for PMP_dias against normal distribution:
Statistic: 0.9472
P-value: 0.0000
The PMP_dias variable does not follow a normal distribution.

II- Scatter Plot for Numeric Variables

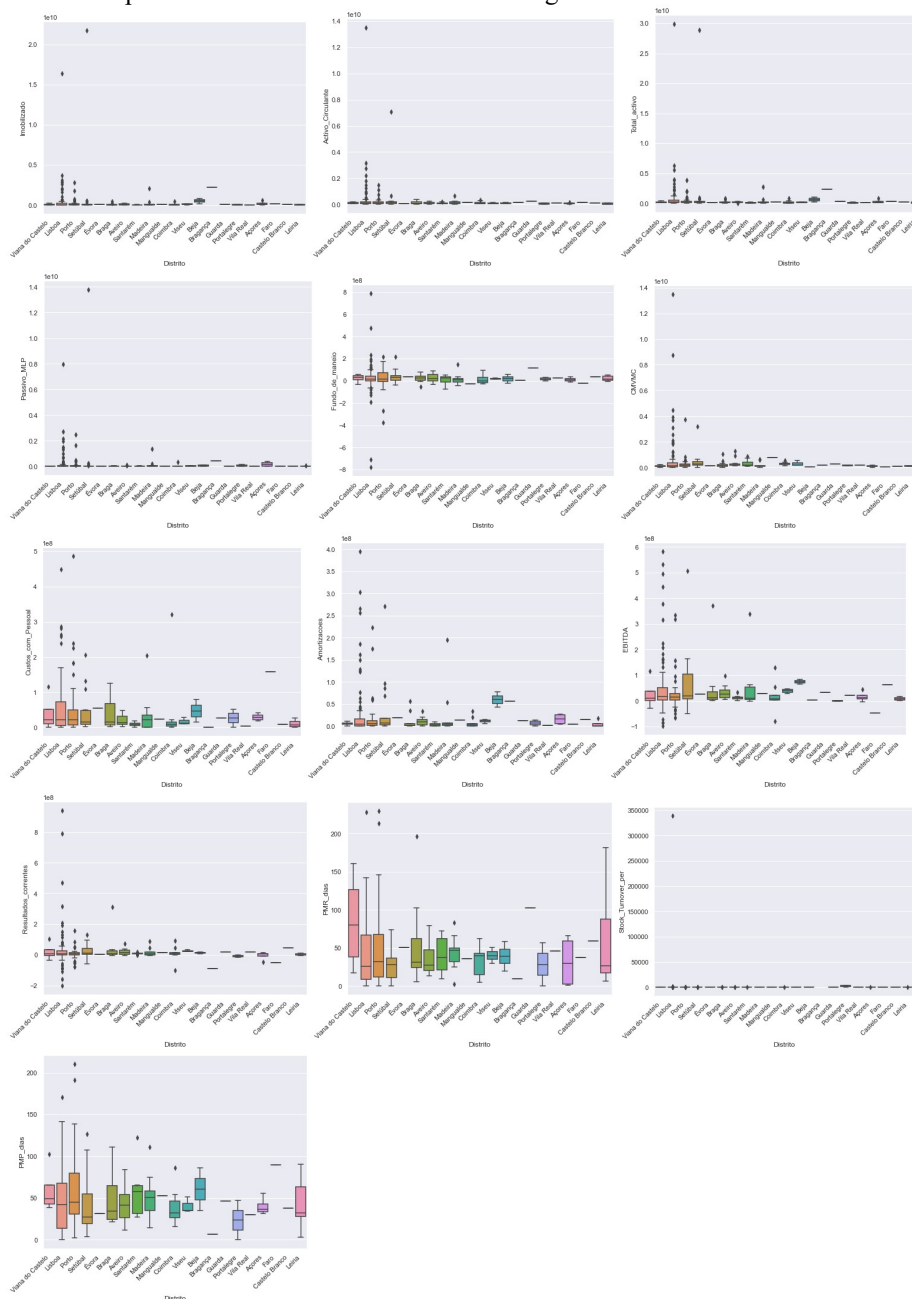


III – Outliers removed from scatter plot

Excluded Observations

('Custos_com_Pessoal', 'Numero_de_empregados'): [17, 18, 68, 70, 73, 79, 80, 82, 131],
 ('Result_Liq', 'EBITDA'): [16, 19, 20, 47, 15, 23, 30, 31, 55, 81, 95, 29, 34, 18, 44, 77, 81, 95],
 ('Result_Liq', 'Resultados_correntes'): [16, 19, 20, 47, 55, 31],
 ('EBITDA', 'Resultados_correntes'): [16, 19, 20, 47, 15, 23, 30, 31, 55, 81, 95, 29, 34, 18,
 44, 77, 81, 95],
 ('Amortizacoes', 'Imobilizado'): [2, 15, 16, 17, 18, 20, 25, 29, 30, 34, 40, 44, 47, 55, 70, 77,
 81, 83, 94, 95, 97, 98, 129, 138, 142, 194, 223, 22, 48, 83, 247],
 ('Total_activo', 'Imobilizado'): [16, 34, 15, 25],
 ('Total_activo', 'Activo_Circulante'): [16, 34, 15, 25],
 ('Capital_Proprio', 'Total_activo'): [16, 34, 15, 25],
 ('Passivo_MLP', 'Imobilizado'): [16, 34, 15, 25, 34].

IV- Boxplots of numerical variables across categories of Distrito



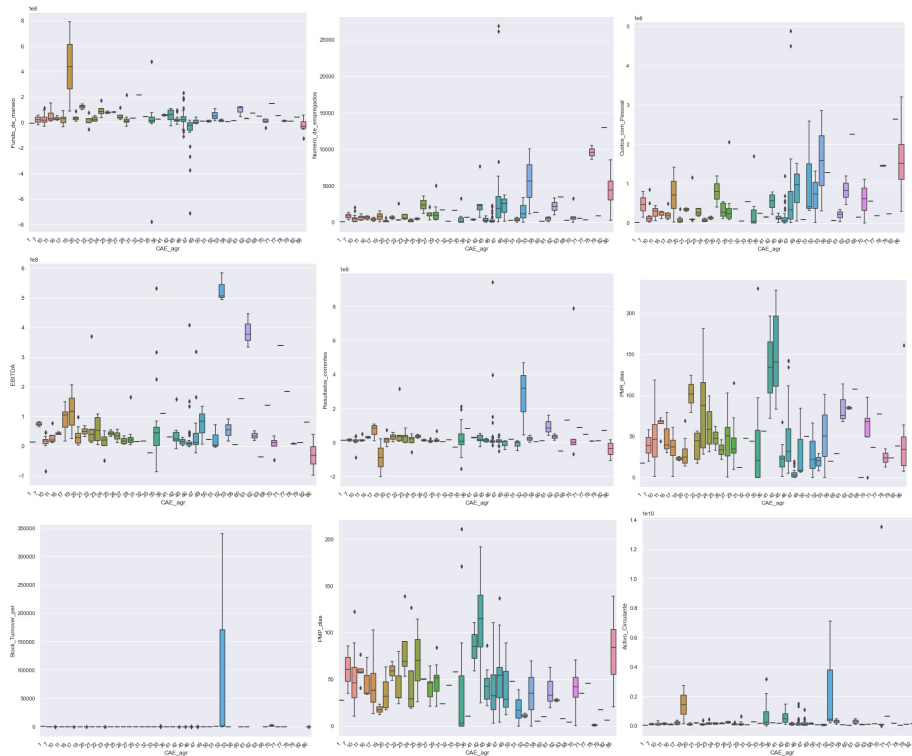
V- SPSS Output of the Independent-Samples Kruskal-Wallis and Median Test for Distrito

Hypothesis Test Summary				
	Null Hypothesis	Test	Sig. ^{a,b}	Decision
1	The medians of Imobilizado are the same across categories of Distrito.	Independent-Samples Median Test	,712	Retain the null hypothesis.
2	The distribution of Imobilizado is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,712	Retain the null hypothesis.
3	The medians of Activo_Circulante are the same across categories of Distrito.	Independent-Samples Median Test	,585	Retain the null hypothesis.
4	The distribution of Activo_Circulante is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,752	Retain the null hypothesis.
5	The medians of Total_activo are the same across categories of Distrito.	Independent-Samples Median Test	,717	Retain the null hypothesis.
6	The distribution of Total_activo is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,679	Retain the null hypothesis.
7	The medians of Capital_Proprio are the same across categories of Distrito.	Independent-Samples Median Test	,385	Retain the null hypothesis.
8	The distribution of Capital_Proprio is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,283	Retain the null hypothesis.
9	The medians of Passivo_MLP are the same across categories of Distrito.	Independent-Samples Median Test	,593	Retain the null hypothesis.
10	The distribution of Passivo_MLP is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,842	Retain the null hypothesis.
11	The medians of Fundo_de_maneio are the same across categories of Distrito.	Independent-Samples Median Test	,930	Retain the null hypothesis.
12	The distribution of Fundo_de_maneio is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,968	Retain the null hypothesis.
13	The medians of Numero_de_empregados are the same across categories of Distrito.	Independent-Samples Median Test	,623	Retain the null hypothesis.
14	The distribution of Numero_de_empregados is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,631	Retain the null hypothesis.
15	The medians of Volume_de_Negocios are the same across categories of Distrito.	Independent-Samples Median Test	,115	Retain the null hypothesis.
16	The distribution of Volume_de_Negocios is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,272	Retain the null hypothesis.
17	The medians of CMVMC are the same across categories of Distrito.	Independent-Samples Median Test	,005	Reject the null hypothesis.
18	The distribution of CMVMC is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,140	Retain the null hypothesis.
19	The medians of Custos_com_Pessoal are the same across categories of Distrito.	Independent-Samples Median Test	,143	Retain the null hypothesis.
20	The distribution of Custos_com_Pessoal is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,407	Retain the null hypothesis.
21	The medians of Amortizacoes are the same across categories of Distrito.	Independent-Samples Median Test	,594	Retain the null hypothesis.
22	The distribution of Amortizacoes is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,526	Retain the null hypothesis.
23	The medians of EBITDA are the same across categories of Distrito.	Independent-Samples Median Test	,360	Retain the null hypothesis.
24	The distribution of EBITDA is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,407	Retain the null hypothesis.
25	The medians of Resultados_correntes are the same across categories of Distrito.	Independent-Samples Median Test	,580	Retain the null hypothesis.
26	The distribution of Resultados_correntes is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,381	Retain the null hypothesis.
27	The medians of Result_Liq are the same across categories of Distrito.	Independent-Samples Median Test	,491	Retain the null hypothesis.
28	The distribution of Result_Liq is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,317	Retain the null hypothesis.
29	The medians of PMR_dias are the same across categories of Distrito.	Independent-Samples Median Test	,853	Retain the null hypothesis.
30	The distribution of PMR_dias is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,933	Retain the null hypothesis.
31	The medians of Stock_Turnover_per are the same across categories of Distrito.	Independent-Samples Median Test	,176	Retain the null hypothesis.
32	The distribution of Stock_Turnover_per is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,444	Retain the null hypothesis.
33	The medians of PMP_dias are the same across categories of Distrito.	Independent-Samples Median Test	,877	Retain the null hypothesis.
34	The distribution of PMP_dias is the same across categories of Distrito.	Independent-Samples Kruskal-Wallis Test	,811	Retain the null hypothesis.

a. The significance level is .050.

b. Asymptotic significance is displayed.

VI- Boxplots of numerical variables across categories of CAE_agr



VII- SPSS Output of the Independent-Samples Kruskal-Wallis and Median Test for CAE_agr

Hypothesis Test Summary				
	Null Hypothesis	Test	Sig. ^{a,b}	Decision
1	The medians of Imobilizado are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
2	The distribution of Imobilizado is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
3	The medians of Activo_Circulante are the same across categories of CAE_agr.	Independent-Samples Median Test	,077	Retain the null hypothesis.
4	The distribution of Activo_Circulante is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	,034	Reject the null hypothesis.
5	The medians of Total_activo are the same across categories of CAE_agr.	Independent-Samples Median Test	,006	Reject the null hypothesis.
6	The distribution of Total_activo is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
7	The medians of Capital_Proprio are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
8	The distribution of Capital_Proprio is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
9	The medians of Passivo_MLP are the same across categories of CAE_agr.	Independent-Samples Median Test	,048	Reject the null hypothesis.
10	The distribution of Passivo_MLP is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	,002	Reject the null hypothesis.
11	The medians of Fundo_de_manejo are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
12	The distribution of Fundo_de_manejo is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
13	The medians of Numero_de_empregados are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
14	The distribution of Numero_de_empregados is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
15	The medians of Volume_de_Negocios are the same across categories of CAE_agr.	Independent-Samples Median Test	,193	Retain the null hypothesis.
16	The distribution of Volume_de_Negocios is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	,108	Retain the null hypothesis.
17	The medians of CMVMC are the same across categories of CAE_agr.	Independent-Samples Median Test	,005	Reject the null hypothesis.
18	The distribution of CMVMC is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
19	The medians of Custos_com_Pessoal are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
20	The distribution of Custos_com_Pessoal is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
21	The medians of Amortizacoes are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
22	The distribution of Amortizacoes is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
23	The medians of EBITDA are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
24	The distribution of EBITDA is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
25	The medians of Resultados_correntes are the same across categories of CAE_agr.	Independent-Samples Median Test	,006	Reject the null hypothesis.
26	The distribution of Resultados_correntes is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
27	The medians of Result_Liq are the same across categories of CAE_agr.	Independent-Samples Median Test	,013	Reject the null hypothesis.
28	The distribution of Result_Liq is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
29	The medians of PMR_dias are the same across categories of CAE_agr.	Independent-Samples Median Test	<,001	Reject the null hypothesis.
30	The distribution of PMR_dias is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
31	The medians of Stock_Turnover_per are the same across categories of CAE_agr.	Independent-Samples Median Test	,002	Reject the null hypothesis.
32	The distribution of Stock_Turnover_per is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.
33	The medians of PMP_dias are the same across categories of CAE_agr.	Independent-Samples Median Test	,008	Reject the null hypothesis.
34	The distribution of PMP_dias is the same across categories of CAE_agr.	Independent-Samples Kruskal-Wallis Test	<,001	Reject the null hypothesis.

a. The significance level is ,050.

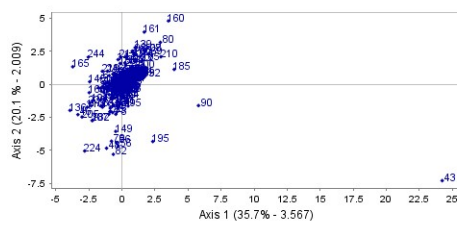
b. Asymptotic significance is displayed.

VIII- Contributions of the active variables to the axes (%)

Label of the variable	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5
EBITDA	14,7	1,6	1,3	3,4	5,3
Passivo_MLP	12,3	0,0	2,4	6,5	8,3
Imobilizado	12,2	0,0	1,6	0,2	1,4
Activo_Circulante	12,2	0,1	3,2	1,9	9,3
Amortizacoes	10,4	0,6	0,9	17,7	4,3
Capital_Proprio	9,1	3,0	0,2	3,3	7,3
Total_activo	7,3	2,5	1,8	4,3	30,8
Resultados_correntes	6,6	12,5	0,1	21,1	2,2
Result_Liq	5,6	12,7	0,1	22,1	3,5
Volume_de_Negocios	4,6	0,8	33,4	0,8	2,6
Custos_com_Pessoal	2,4	25,2	3,9	1,6	0,1
Numero_de_empregados	2,4	27,9	3,2	3,5	0,0
CMVMC	0,2	0,8	47,7	0,4	1,3
Fundo_de_manejo	0,0	12,2	0,3	13,2	23,6

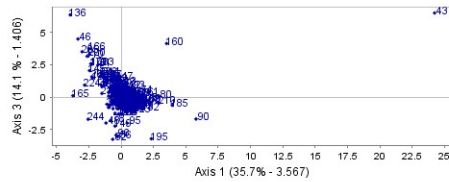
IX- Summary of the PCA of the ratio variables including case 43

Cases - Axes 1 & 2



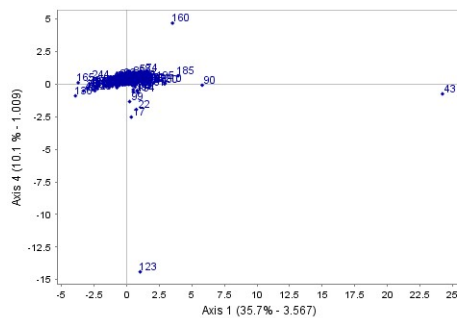
◆ Active cases

Cases - Axes 1 & 3



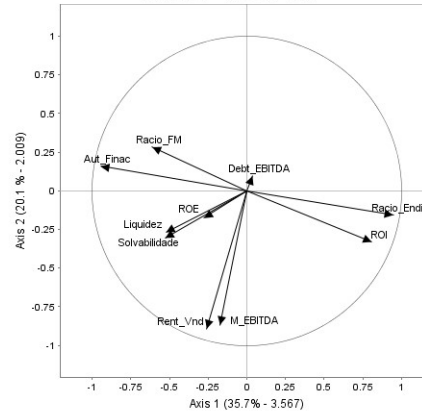
◆ Active cases

Cases - Axes 1 & 4



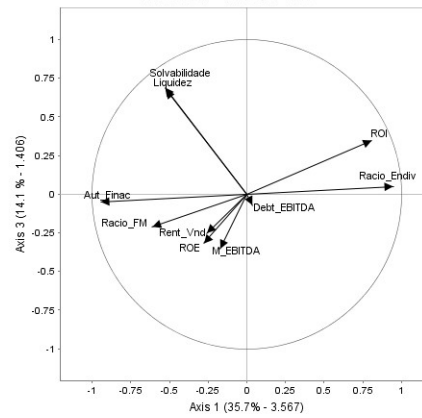
◆ Active cases

Variables - Axes 1 & 2

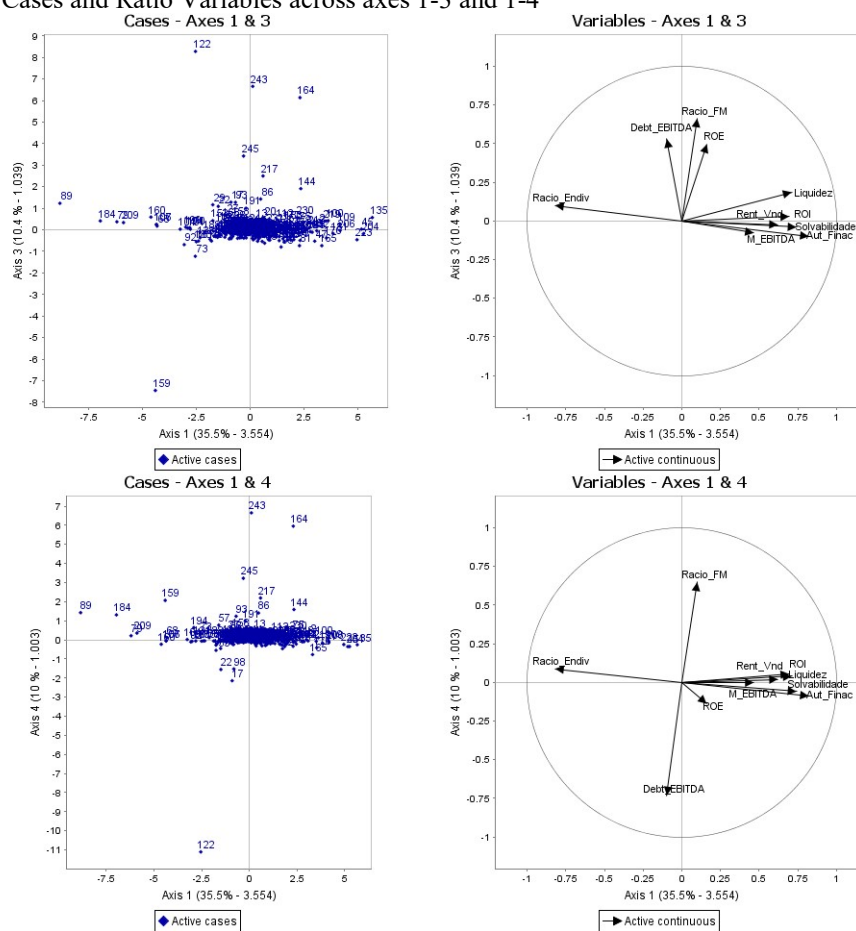


→ Active continuous

Variables - Axes 1 & 3



X- Cases and Ratio Variables across axes 1-3 and 1-4



XI – Boxplots for economic and financial variables, by cluster

